

SGA 3, Exposé XXI
Root Data
English Reconstruction

M. Demazure

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Exposé XXI. Root Data

This Exposé¹ collects, in the absence of a suitable reference,² known results on root data (that is, “abstract” root systems), most of which will be used below.

Notation. We denote by $\mathbb{Q}_+$ the set of positive (or zero) rational numbers; $\mathbb{Z} \cap \mathbb{Q}_+ = \mathbb{N}$. Let V be a $\mathbb{Q}$ -vector space. If A (respectively B) is a subset of $\mathbb{Q}$ (respectively of V), we denote by $A \cdot B$ the image of $A \otimes B$ under the morphism $\mathbb{Q} \otimes V \rightarrow V$, in other words the set of linear combinations of elements of B with coefficients in A . We write $-B = \{-1\}B$. We denote by $E - F$ the set of elements of E that do not belong to F .

1 Generalities

1.1 Definitions and first properties

Definition 1.1.1. Let M and M^* be two free finite-type $\mathbb{Z}$ -modules in duality. Write

$$V = M \otimes \mathbb{Q}, \quad V^* = M^* \otimes \mathbb{Q};$$

these are two $\mathbb{Q}$ -vector spaces in duality. We identify M (respectively M^*) with a subset of V (respectively V^*). The canonical bilinear form on $M^* \times M$ (respectively $V^* \times V$) is denoted $(,)$.

Let R be a finite subset of M . Suppose given a map $\alpha \mapsto \alpha^*$ from R to M^* ; the set of α^* , for $\alpha \in R$, is denoted R^* . To every $\alpha \in R$ associate the endomorphism s_α (respectively s_α^*) of M and V (respectively of M^* and V^*) given by

$$\begin{aligned} (1) \quad s_\alpha(x) &= x - (\alpha^*, x)\alpha, & s_\alpha &= \text{id} - \alpha^* \otimes \alpha, \\ (1^*) \quad s_\alpha^*(u) &= u - (u, \alpha)\alpha^*, & s_\alpha^* &= \text{id} - \alpha \otimes \alpha^*. \end{aligned}$$

The pair (R, R^*) (more precisely, the pair $(R, R \rightarrow M^*)$) is called a *root datum* in (M, M^*) , and (M, M^*, R, R^*) is called a root datum, if the following axioms hold:

$$\begin{aligned} (\text{DR I}) \quad (\alpha^*, \alpha) &= 2 & \text{for every } \alpha \in R, \\ (\text{DR II}) \quad s_\alpha(R) &\subset R, \quad s_\alpha^*(R^*) \subset R^* & \text{for every } \alpha \in R. \end{aligned}$$

The set R is called the *root system* of the root datum $\mathcal{R} = (M, M^*, R, R^*)$. The elements of R (respectively R^*) are called the *roots* (respectively *coroots*) of the root datum.

Remark 1.1.2. Axiom (DR I) is equivalent to any one of the following properties:

$$\begin{aligned} (2, 2^*) \quad s_\alpha s_\alpha &= \text{id}, & s_\alpha^* s_\alpha^* &= \text{id}, \\ (3, 3^*) \quad s_\alpha(\alpha) &= -\alpha, & s_\alpha^*(\alpha^*) &= -\alpha^*. \end{aligned}$$

¹Editors' note: Version of 13 October 2024.

²Editors' note: For results on root systems (Sections 1–5), one may consult *Groupes et algèbres de Lie*, Chapter VI.

Remark 1.1.3. Axioms (DR I) and (DR II) imply

$$R = -R, \quad R^* = -R^*, \quad 0 \notin R, \quad 0 \notin R^*.$$

Lemma 1.1.4. *The map $R \rightarrow R^*$ is a bijection. More generally, if $\alpha, \beta \in R$ and $(\alpha^*, x) = (\beta^*, x)$ for every $x \in R$, then $\alpha = \beta$.*

Proof. Indeed, one then has

$$s_\beta(\alpha) = \alpha - 2\beta, \quad s_\alpha(\beta) = \beta - 2\alpha.$$

It follows at once that

$$s_\beta s_\alpha(\alpha) = 2\beta - \alpha = \alpha + 2(\beta - \alpha), \quad s_\beta s_\alpha(\beta - \alpha) = s_\beta(\beta - \alpha) = \beta - \alpha.$$

Hence, by (DR II),

$$(s_\beta s_\alpha)^n(\alpha) = \alpha + 2n(\beta - \alpha) \in R.$$

Since R is finite, $\beta - \alpha = 0$. □

Corollary 1.1.5. *The inverse map $R^* \rightarrow R$ defines a root datum*

$$\mathcal{R}^* = (M^*, M, R^*, R),$$

*called the dual of $\mathcal{R}$.*³

Definition 1.1.6. We denote by $\Gamma_0(R)$ the subgroup of M generated by R . We denote by $\mathcal{V}(R)$ the vector subspace of V generated by R , that is,

$$\mathcal{V}(R) = \Gamma_0(R) \otimes \mathbb{Q}.$$

Applying these definitions to $\mathcal{R}^*$, one likewise obtains $\Gamma_0(R^*)$ and $\mathcal{V}(R^*)$.

The *reductive rank* of $\mathcal{R}$ is the number

$$\text{rgred}(\mathcal{R}) = \text{rank}(M) = \dim(V) = \dim(V^*) = \text{rank}(M^*) = \text{rgred}(\mathcal{R}^*).$$

The *semisimple rank* of $\mathcal{R}$ is the number

$$\text{rgss}(\mathcal{R}) = \text{rank}(R) = \text{rank}(\Gamma_0(R)) = \dim(\mathcal{V}(R)).$$

Thus $\text{rgss}(\mathcal{R}) \leq \text{rgred}(\mathcal{R})$.

We shall see below that $\text{rgss}(\mathcal{R}) = \text{rgss}(\mathcal{R}^*)$, that is, that $\mathcal{V}(R)$ and $\mathcal{V}(R^*)$ have the same dimension.

Definition 1.1.7.

The root datum $\mathcal{R}$ is called *semisimple* (respectively *trivial*) if $\text{rgss}(\mathcal{R}) = \text{rgred}(\mathcal{R})$ (respectively $\text{rgss}(\mathcal{R}) = 0$). Thus $\mathcal{R}$ is trivial if and only if R is empty. The trivial root datum of reductive rank zero is denoted

$$0 = (\{0\}, \{0\}, \emptyset, \emptyset).$$

³Editors' note: $(\alpha^*)^* = \alpha$.

Definition 1.1.8. We denote by $W(\mathcal{R})$ the group of transformations of M generated by the s_α , $\alpha \in R$. It is called the *Weyl group* of $\mathcal{R}$. We write

$$W^*(\mathcal{R}) = W(\mathcal{R}^*).$$

Then $W(\mathcal{R})$ acts on R , $\Gamma_0(R)$, $\mathcal{V}(R)$, M , and V . If $w \in W(\mathcal{R})$ and $x \in M$ (respectively $x \in V$), then

$$wx - x \in \Gamma_0(R) \quad (\text{respectively } wx - x \in \mathcal{V}(R));$$

this follows immediately from formula (1). Likewise for $W^*(\mathcal{R})$.

Lemma 1.1.9. *For every $\alpha \in R$, $x \in V$, and $u \in V^*$, one has*

$$(4) \quad (s_\alpha^*(u), s_\alpha(x)) = (u, x).$$

Proof. Taking the scalar product of (1) and (1*), the left-hand side is

$$(u, x) + (u, \alpha)(\alpha^*, x)((\alpha^*, \alpha) - 2) = (u, x).$$

□

Remark 1.1.10. If one assumes $0 \notin R$ and $0 \notin R^*$, then Lemma 1.1.9 is equivalent to (DRI).

Corollary 1.1.11. *Let $h \mapsto h^\vee$ denote the isomorphism from $\text{GL}(M)$ onto $\text{GL}(M^*)$ that associates to h its contragredient. Formula (4) may then also be written*

$$(5) \quad (s_\alpha)^\vee = s_\alpha^*.$$

Corollary 1.1.12. *The preceding isomorphism induces an isomorphism from $W(\mathcal{R})$ onto $W^*(\mathcal{R})$.*

Scholium 1.1.13. *By the preceding result, we shall identify W and W^* , and regard W as a group of transformations of*

$$R, R^*, M, M^*, \Gamma_0(R), \Gamma_0(R^*), V, V^*, \mathcal{V}(R), \mathcal{V}(R^*).$$

We shall write s_α for s_α^ .*

1.2 The map p

Lemma 1.2.1. *Let $p : M \rightarrow M^*$ (respectively $V \rightarrow V^*$) be the linear map defined by*

$$(6) \quad p(x) = \sum_{u \in R^*} (u, x)u.$$

Write $\ell(x) = (p(x), x)$. Then:

$$\begin{aligned} (7) \quad & \ell(x) \geq 0, & \ell(\alpha) > 0 & & \text{for } \alpha \in R, \\ (8) \quad & \ell(wx) = \ell(x) & & & \text{for } w \in W, \\ (9) \quad & (p(x), y) = (p(y), x) & & & \text{for } x, y \in V, \\ (10) \quad & \ell(\alpha)\alpha^* = 2p(\alpha) & & & \text{for } \alpha \in R. \end{aligned}$$

Proof. The first three relations are immediate.⁴ We prove the last one. By (1), for $u \in V^*$,

$$\begin{aligned} (u, \alpha)^2 \alpha^* &= (u, \alpha)u - (u, \alpha)s_\alpha(u) \\ &= (u, \alpha)u + (u, s_\alpha(\alpha))s_\alpha(u) \\ &= (u, \alpha)u + (s_\alpha(u), \alpha)s_\alpha(u) \quad (s_\alpha^2 = \text{id}). \end{aligned}$$

Since s_α permutes R^* by (DR II), summing over $u \in R^*$ gives the result. $\square$

Scholium 1.2.2. *Relation (10) says that the map $\alpha \mapsto \ell(\alpha)\alpha^*$ is the restriction to R of a linear map from M to M^* . In particular,*

$$(-\alpha)^* = -\alpha^*.$$

Corollary 1.2.3. *The map p induces an isomorphism from $\mathcal{V}(R)$ onto $\mathcal{V}(R^*)$.*

Proof. The map p sends $\mathcal{V}(R)$ onto $\mathcal{V}(R^*)$. Hence

$$\dim \mathcal{V}(R) \geq \dim \mathcal{V}(R^*).$$

Applying this inequality to the dual root datum gives

$$\dim \mathcal{V}(R) = \dim \mathcal{V}(R^*),$$

so the surjective map p is also bijective. $\square$

Corollary 1.2.4. *One has $\dim \mathcal{V}(R) = \dim \mathcal{V}(R^*)$, and therefore*

$$\text{rgss}(\mathcal{R}) = \text{rgss}(\mathcal{R}^*).$$

Corollary 1.2.5. *The bilinear form (u, x) is nondegenerate on $\mathcal{V}(R^*) \times \mathcal{V}(R)$, and thus puts these $\mathbb{Q}$ -vector spaces in duality.*

Proof. If $(u, x) = 0$ for every $u \in R^*$, then $p(x) = 0$. $\square$

Corollary 1.2.6. *The symmetric bilinear form $(p(x), y)$ is positive and nondegenerate on $\mathcal{V}(R)$.*

Corollary 1.2.7. *The group W acts faithfully on R (and hence on the other sets listed in Scholium 1.1.13).*

Proof. Let $u \in V^*$, and suppose $w \in W$ satisfies $w(\alpha) = \alpha$ for every $\alpha \in R$. We prove $w(u) = u$. For every $\alpha \in R$,

$$(w(u) - u, \alpha) = (w(u), \alpha) - (u, \alpha) = (u, w^{-1}(\alpha)) - (u, \alpha) = 0.$$

But $w(u) - u \in \mathcal{V}(R^*)$. Since it is orthogonal to every root, it vanishes by Corollary 1.2.5. $\square$

Corollary 1.2.8. *The group W is finite.*

Proposition 1.2.9. *The operations of W respect the correspondence between roots and coroots. In other words, for $\alpha \in R$ and $w \in W$,*

$$w(\alpha^*) = w(\alpha)^*.$$

⁴Editors' note: Formula (8) follows from (4), (2,2*), and (DR II).

Proof. It suffices to verify this for $w = s_\beta$, $\beta \in R$, that is, to verify

$$s_\beta(\alpha^*) = s_\beta(\alpha)^*.$$

Now $\ell(s_\beta(\alpha))s_\beta(\alpha)^*/2$ equals

$$\begin{aligned} p(s_\beta(\alpha)) &= \sum_{u \in R^*} (u, s_\beta(\alpha))u \\ &= \sum_{u \in R^*} (s_\beta(u), \alpha)u \\ &= \sum_{u \in R^*} (u, \alpha)s_\beta(u) = s_\beta(p(\alpha)). \end{aligned}$$

Since $\ell(s_\beta(\alpha)) = \ell(\alpha)$, we obtain

$$s_\beta(\alpha)^* = s_\beta\left(\frac{2p(\alpha)}{\ell(\alpha)}\right) = s_\beta(\alpha^*).$$

□

Corollary 1.2.10. *If $\alpha \in R$ and $w \in W$, then*

$$ws_\alpha w^{-1} = s_{w(\alpha)}.$$

Proof. For $x \in V$,

$$ws_\alpha w^{-1}(x) = x - (\alpha^*, w^{-1}(x))w(\alpha) = x - (w(\alpha^*), x)w(\alpha).$$

By Proposition 1.2.9, the last expression is

$$x - (w(\alpha)^*, x)w(\alpha) = s_{w(\alpha)}(x).$$

□

Corollary 1.2.11.

Let $R' \subset R$. The quadruple (M, M^, R', R'^*) is a root datum if and only if $\alpha, \beta \in R'$ implies $s_\alpha(\beta) \in R'$.*

2 Relations between two roots

2.1 Proportional roots

Proposition 2.1.1. *Let α and β be two roots. The following conditions are equivalent:*

- (i) *there exists $k \in \mathbb{Q}$ such that $\alpha = k\beta$;*
- (ii) *$s_\alpha = s_\beta$.*

Under these conditions, $\alpha^ = k^{-1}\beta^*$, and k is one of*

$$1, -1, 2, -2, \frac{1}{2}, -\frac{1}{2}.$$

Proof. Assume first (i). Then

$$\alpha^* = \ell(\alpha)^{-1} 2p(\alpha) = k^{-2} \ell(\beta)^{-1} 2k p(\beta) = k^{-1} \beta^*.$$

This immediately gives $s_\alpha = s_\beta$. Conversely, if $s_\alpha = s_\beta$, then

$$\alpha = s_\alpha(-\alpha) = s_\beta(-\alpha) = (\beta^*, \alpha)\beta - \alpha,$$

so

$$2\alpha = (\beta^*, \alpha)\beta, \quad (\beta^*, \alpha) \in \mathbb{Z};$$

thus (ii) implies (i). Finally, if $\alpha = k\beta$, then $\alpha^* = k^{-1}\beta^*$, whence

$$(\alpha^*, \beta) = 2k^{-1}, \quad (\beta^*, \alpha) = 2k.$$

Thus $2k$ and $2k^{-1}$ are integers, which gives the stated list. $\square$

Application 2.1.2. Root data $\mathcal{R}$ with $\text{rgss}(\mathcal{R}) = 1$ are of one of the following two types:

- (i) Type A_1 : there exists $\alpha \in M$ such that the roots are α and $-\alpha$. The coroots are then α^* and $-\alpha^*$.
- (ii) Type A'_1 : there exists $\alpha \in M$ such that the roots are $\alpha, -\alpha, 2\alpha, -2\alpha$. The coroots are then $\alpha^*, -\alpha^*, \alpha^*/2, -\alpha^*/2$.

Definition 2.1.3. A root $\alpha \in R$ is called *indivisible* if $\alpha/2 \notin R$. The root datum $\mathcal{R}$ is called *reduced* if every root is indivisible.

The root datum $\mathcal{R}$ is reduced if and only if $\mathcal{R}^*$ is reduced. If α is indivisible and $w \in W$, then $w(\alpha)$ is indivisible.

Definition 2.1.4. Let $\alpha \in R$. If α is indivisible, set $\text{ind}(\alpha) = \alpha$; otherwise set $\text{ind}(\alpha) = \alpha/2$.

Corollary 2.1.5. If $\alpha \in R$ is indivisible and $k\alpha \in R$, where $k \in \mathbb{Q}$, then $k \in \mathbb{Z}$.

Proposition 2.1.6. Let $\mathcal{R}$ be a root datum. Then

$$\text{ind}(\mathcal{R}) = (M, M^*, \text{ind}(R), \text{ind}(R)^*)$$

is a reduced root datum, and

$$W(\text{ind}(\mathcal{R})) = W(\mathcal{R}).$$

Proof. The quadruple $\text{ind}(\mathcal{R})$ is a root datum by Corollary 1.2.11, since the Weyl group permutes the indivisible roots. The second assertion follows from Proposition 2.1.1. $\square$

Remark 2.1.7. If $\mathcal{R}$ is not reduced, then

$$\text{ind}(R)^* \neq \text{ind}(R^*), \quad \text{ind}(\mathcal{R}^*) \neq \text{ind}(\mathcal{R})^*.$$

2.2 Orthogonal roots

Lemma 2.2.1. *Let α and β be two roots. Then*

$$(11) \quad \ell(\alpha)(\alpha^*, \beta) = \ell(\beta)(\beta^*, \alpha).$$

Proof. This follows immediately from Lemma 1.2.1, formulas (9) and (10). $\square$

Corollary 2.2.2. *Let $\alpha, \beta \in R$. The following conditions are equivalent:*

- (i) $(\alpha^*, \beta) = 0$;
- (i bis) $(\beta^*, \alpha) = 0$;
- (ii) $(p(\alpha), \beta) = 0$;
- (iii) $s_\alpha(\beta) = \beta$;
- (iii bis) $s_\beta(\alpha) = \alpha$;
- (iv) $s_\alpha(\beta^*) = \beta^*$;
- (iv bis) $s_\beta(\alpha^*) = \alpha^*$;
- (v) $s_\alpha \neq s_\beta$, and s_α and s_β commute.

Proof. All the equivalences are immediate except those involving (v). We first show that (i), together with its equivalent condition (i bis), implies (v). One has

$$s_\alpha s_\beta(x) = x - (\beta^*, x)\beta - (\alpha^*, x)\alpha + (\beta^*, x)(\alpha^*, \beta)\alpha.$$

If $(\alpha^*, \beta) = 0$, then $(\beta^*, \alpha) = 0$, and consequently

$$s_\alpha s_\beta(x) = s_\beta s_\alpha(x).$$

Conversely, suppose (v). Then

$$s_\alpha = s_\beta s_\alpha s_\beta = s_{s_\beta(\alpha)} \quad \text{by Corollary 1.2.10.}$$

By Proposition 2.1.1, there is a $k \in \mathbb{Q}^*$ such that

$$\alpha = k s_\beta(\alpha) = k\alpha - k(\beta^*, \alpha)\beta.$$

Since $s_\alpha \neq s_\beta$, the roots α and β are not proportional, again by Proposition 2.1.1. It follows that $(\beta^*, \alpha) = 0$. $\square$

Definition 2.2.3. Two roots satisfying the equivalent conditions of Corollary 2.2.2 are called *orthogonal*.

Remark 2.2.4. The roots α and β are orthogonal if and only if the coroots α^* and β^* are orthogonal.

Lemma 2.2.5. *If α and β are two orthogonal roots, then $\alpha + \beta \in R$ if and only if $\alpha - \beta \in R$.*

Proof. Indeed,

$$s_\beta(\alpha + \beta) = s_\beta(\alpha) + s_\beta(\beta) = \alpha - \beta.$$

$\square$

Lemma 2.2.6. *Let α and β be two nonorthogonal roots. For a coroot γ^* , define $\ell(\gamma^*)$ to be the value of ℓ on the root γ^* of $\mathcal{R}^*$. Then*

$$(12) \quad \ell(\alpha)\ell(\alpha^*) = \ell(\beta)\ell(\beta^*).$$

Proof. Multiply formula (11) by the corresponding formula for $\mathcal{R}^*$, and use $(\gamma^*)^* = \gamma$ for every $\gamma \in R$. This gives

$$(\beta^*, \alpha)(\alpha^*, \beta)\ell(\alpha)\ell(\alpha^*) = (\beta^*, \alpha)(\alpha^*, \beta)\ell(\beta)\ell(\beta^*).$$

The common integer factor is nonzero because the roots are not orthogonal, so it may be cancelled. $\square$

2.3 General case

Proposition 2.3.1. *If α and β are any two roots, then*

$$(13) \quad 0 \leq (\alpha^*, \beta)(\beta^*, \alpha) \leq 4.$$

If α and β are neither proportional nor orthogonal, then

$$1 \leq (\alpha^*, \beta)(\beta^*, \alpha) \leq 3.$$

Proof. One has

$$4(p(\alpha), \beta)^2 = \ell(\alpha)\ell(\beta)(\alpha^*, \beta)(\beta^*, \alpha).$$

On the other hand, by Corollary 1.2.6, the symmetric bilinear form $(p(x), y)$ is positive and nondegenerate on $\mathcal{V}(R)$. Hence

$$(p(x), y)^2 \leq \ell(x)\ell(y).$$

⁵ The first inequality follows. In the nonproportional, nonorthogonal case, the product is a nonzero integer and cannot equal 4, which gives the second inequality. $\square$

Corollary 2.3.2. *Let α and β be two nonorthogonal roots. If $\ell(\alpha) = \ell(\beta)$, then there is a $w \in W$ such that $w(\alpha) = \beta$.*

Proof. If α and β are proportional, equality of their ℓ -values gives $\alpha = \beta$ or $\alpha = -\beta$; take $w = 1$ or $w = s_\alpha$, respectively. If they are neither proportional nor orthogonal, formula (11) and Proposition 2.3.1 give

$$(\beta^*, \alpha) = (\alpha^*, \beta) = \pm 1.$$

If both integers are 1, take $w = s_\beta s_\alpha s_\beta$. If both are -1 , take $w = s_\alpha s_\beta$. $\square$

Corollary 2.3.3. *Let α and β be roots. If $\alpha \neq \beta$ and $(\beta^*, \alpha) > 0$, then $\alpha - \beta$ is a root. Respectively, if $\alpha \neq -\beta$ and $(\beta^*, \alpha) < 0$, then $\alpha + \beta$ is a root.*

Proof. The second case follows from the first by replacing β by $-\beta$. Suppose first that α and β are proportional and that $(\beta^*, \alpha) > 0$. Then

$$\beta = \alpha, \quad 2\beta = \alpha, \quad \text{or} \quad \beta = 2\alpha.$$

⁵Editors' note: equality holds if and only if x and y are proportional.

The first case is excluded. In the other two cases, $\alpha - \beta = \beta \in R$ and $\alpha - \beta = -\alpha \in R$, respectively.

If α and β are not proportional, then (α^*, β) and (β^*, α) are strictly positive integers whose product is at most 3. Thus one of them equals 1. If $(\beta^*, \alpha) = 1$, then

$$\alpha - \beta = s_\beta(\alpha) \in R;$$

if $(\alpha^*, \beta) = 1$, then

$$\alpha - \beta = -s_\alpha(\beta) \in R.$$

□

Lemma 2.3.4. *Let α and β be two nonproportional roots. If $\beta - \alpha$ is not a root, then $\beta + k\alpha \in R$ for $k = -(\alpha^*, \beta)$, but not for $k = -(\alpha^*, \beta) + 1$.*

Proof. Indeed,

$$\beta - (\alpha^*, \beta)\alpha = s_\alpha(\beta) \in R,$$

whereas

$$\beta + (-(\alpha^*, \beta) + 1)\alpha = s_\alpha(\beta - \alpha) \notin R.$$

□

Proposition 2.3.5. *Let α and β be two nonproportional roots. The set of integers k for which $\beta + k\alpha \in R$ is an interval $[p, q]$, with $p \leq 0$ and $q \geq 0$, and*

$$p + q = -(\alpha^*, \beta).$$

Proof. For the first assertion it suffices, for example, to show that if $\beta + k\alpha \in R$ and $k > 0$ is an integer, then $\beta + (k - 1)\alpha \in R$. This is trivial if $k = 1$. If $k \geq 2$, then

$$(\alpha^*, \beta + k\alpha) = (\alpha^*, \beta) + 2k \geq -3 + 4 > 0,$$

and the assertion follows from Corollary 2.3.3.

Let $[p, q]$ be the interval in question. Applying Lemma 2.3.4 to $\beta + p\alpha$, one obtains

$$q - p = -(\alpha^*, \beta + p\alpha) = -(\alpha^*, \beta) - 2p.$$

⁶ The asserted formula follows. □

Remark 2.3.6. The preceding formula contains the qualitative statements of Lemma 2.2.5 and Corollary 2.3.3.

Complements 2.3.7. ⁷ *By formulas (9) and (10), and by Corollary 1.2.6, the bilinear form on $\mathcal{V}(R)$ defined by*

$$\langle x, y \rangle = (p(x), y)$$

is symmetric and positive definite. For every $\alpha \in R$ and $y \in \mathcal{V}(R)$,

$$\langle \alpha, y \rangle = \frac{\ell(\alpha)}{2}(\alpha^*, y), \quad \ell(\alpha) = \langle \alpha, \alpha \rangle.$$

Consequently, Corollary 2.3.3 gives the following corollary.

Corollary. *Let $\alpha \neq \beta$ in R . If $\alpha - \beta \notin R$, then*

$$\langle \alpha, \beta \rangle \leq 0.$$

⁶Editors' note: in this equality, $+2p$ was corrected to $-2p$.

⁷Editors' note: these complements, useful for the proof of Proposition 3.1.5, were added by the editors.

3 Simple roots, positive roots

3.1 Systems of simple roots

Lemma 3.1.1. *Let α and α_i , $i = 1, \dots, n$, be roots, and suppose that α is distinct from the α_i . If*

$$\alpha = \sum_{i=1}^n q_i \alpha_i, \quad q_i \in \mathbb{Q}_+,$$

then there is an index i such that

$$q_i \neq 0, \quad (\alpha^*, \alpha_i) > 0, \quad \alpha - \alpha_i \in R.$$

Proof. Write

$$2 = (\alpha^*, \alpha) = \sum_{i=1}^n q_i (\alpha^*, \alpha_i).$$

This proves the first two assertions. The third then follows from Corollary 2.3.3. $\square$

Proposition 3.1.2. *Let α and α_i , $i = 1, \dots, n$, be roots. If*

$$\alpha = \sum_{i=1}^n m_i \alpha_i, \quad m_i \in \mathbb{N},$$

then there is a sequence $\beta_1, \dots, \beta_m$ of roots chosen from among the α_i , not necessarily pairwise distinct, such that, on setting

$$\gamma_p = \sum_{i=1}^p \beta_i, \quad p = 1, \dots, m,$$

one has $\gamma_p \in R$ and $\gamma_m = \alpha$.

Proof. We argue by induction on the integer $m' = \sum_i m_i$. If α equals one of the α_i , say α_{i_0} , which necessarily happens when $m' = 1$, take

$$m = 1, \quad \beta_1 = \alpha_{i_0}.$$

Otherwise, Lemma 3.1.1 gives an index i for which $m_i \neq 0$ and $\alpha - \alpha_i$ is a root. Then $m_i - 1 \in \mathbb{N}$ and

$$\alpha - \alpha_i = (m_i - 1)\alpha_i + \sum_{j \neq i} m_j \alpha_j.$$

Apply the induction hypothesis to $\alpha - \alpha_i$, and append α_i to the resulting sequence. $\square$

Corollary 3.1.3. *Let $R' \subset R$. The following conditions are equivalent:*

- (i) *if $\alpha, \beta \in R'$ and $\alpha + \beta \in R$, then $\alpha + \beta \in R'$;*
- (ii) *$(\mathbb{N} \cdot R') \cap R \subset R'$.*

Proof. Clearly (ii) implies (i). The converse follows immediately from Proposition 3.1.2. $\square$

Definition 3.1.4. A set of roots satisfying the equivalent conditions of Corollary 3.1.3 is called *closed*.

Proposition 3.1.5. *Let $\Delta \subset R$ be a set of roots. The following conditions are equivalent:*

(i) the elements of Δ are indivisible and linearly independent, and

$$R \subset (\mathbb{Q}_+ \cdot \Delta) \cup (-\mathbb{Q}_+ \cdot \Delta);$$

(ii) the elements of Δ are linearly independent, and

$$R \subset (\mathbb{N} \cdot \Delta) \cup (-\mathbb{N} \cdot \Delta);$$

(iii) every root can be written uniquely as a linear combination of the elements of Δ , with integer coefficients all of the same sign.

Proof. ⁸ It is clear that (ii) implies (iii). Write $\alpha_1, \dots, \alpha_n$ for the distinct elements of Δ .

Assume (i), and let $\alpha \in R$. There is a unique expression

$$\varepsilon\alpha = \sum_i q_i \alpha_i, \quad q_i \in \mathbb{Q}_+, \quad \varepsilon = \pm 1.$$

We show that the q_i are integers. This is immediate when all but one of them vanish, by Corollary 2.1.5. Otherwise, $\varepsilon\alpha$ is distinct from the α_i , and Lemma 3.1.1 gives an i_0 such that

$$\alpha' = \varepsilon\alpha - \alpha_{i_0} \in R, \quad q_{i_0} \neq 0.$$

Thus

$$\alpha' = (q_{i_0} - 1)\alpha_{i_0} + \sum_{i \neq i_0} q_i \alpha_i.$$

At least one q_i , with $i \neq i_0$, is nonzero, so (i) forces $q_{i_0} - 1 \geq 0$. Repeating this operation with α' , after finitely many steps we find that all q_i are integers. This proves (ii).

Assume now (iii). We first prove that $\alpha_1, \dots, \alpha_n$ are linearly independent over $\mathbb{Q}$. Otherwise there would be an equality

$$x = \sum_{i \in I} a_i \alpha_i = \sum_{j \in J} b_j \alpha_j,$$

where I and J are disjoint subsets of $\{1, \dots, n\}$, at least one of them, say I , is nonempty, and $a_i, b_j \in \mathbb{N}^*$. By the corollary in Complements 2.3.7, one has $\langle \alpha_i, \alpha_j \rangle \leq 0$ for $i \in I$ and $j \in J$. Therefore $\langle x, x \rangle \leq 0$, and hence $x = 0$.

Choose $i_0 \in I$. The two expressions

$$\alpha_{i_0} = \alpha_{i_0} + \sum_{i \in I} a_i \alpha_i = \alpha_{i_0} + \sum_{j \in J} b_j \alpha_j$$

then imply, by (iii), that $a_i = 0 = b_j$ for all i, j , a contradiction. Thus the elements of Δ are linearly independent over $\mathbb{Q}$.

It remains to show that they are indivisible. Suppose that $\beta \in \Delta$ is divisible. Then $\beta/2 \in R$, so

$$\frac{\beta}{2} = \varepsilon \sum_{\alpha \in \Delta} m_\alpha \alpha, \quad m_\alpha \in \mathbb{N}, \quad \varepsilon = \pm 1.$$

Consequently,

$$\beta = 2\varepsilon \sum_{\alpha \in \Delta} m_\alpha \alpha.$$

Uniqueness gives $m_\alpha = 0$ for $\alpha \neq \beta$ and $2\varepsilon m_\beta = 1$, a contradiction. $\square$

⁸Editors' note: in the argument that follows, the order was changed and the proof of the implication (iii) $\Rightarrow$ (i) was spelled out.

Definition 3.1.6. A set Δ of roots satisfying the conditions of Proposition 3.1.5 is called a *system of simple roots*, or a *base* of $\mathcal{R}$.

If $w \in W$ and Δ is a system of simple roots, then $w(\Delta)$ is a system of simple roots.

Remark 3.1.7. This definition involves only R , not $\mathcal{R}$; indeed, it involves only $\text{ind}(R)$.

Remark 3.1.8. If Δ is a system of simple roots, then Δ is a basis of the free abelian group $\Gamma_0(R)$. Hence

$$\text{Card}(\Delta) = \text{rgss}(\mathcal{R}).$$

Remark 3.1.9. The conditions of Proposition 3.1.5 are therefore equivalent to the following:

(i') the elements of Δ are indivisible, their number is at most $\text{rgss}(\mathcal{R})$, and

$$R \subset (\mathbb{Q}_+ \cdot \Delta) \cup (-\mathbb{Q}_+ \cdot \Delta);$$

(ii') $\text{Card}(\Delta) \leq \text{rgss}(\mathcal{R})$, and

$$R \subset (\mathbb{N} \cdot \Delta) \cup (-\mathbb{N} \cdot \Delta).$$

Corollary 3.1.10. *If Δ is a system of simple roots, then $\text{ind}(\Delta^*)$ is a system of simple coroots (i.e. a system of simple roots of $\mathcal{R}^*$).*

Proof. If

$$\beta = \sum_{\alpha \in \Delta} a_\alpha \alpha, \quad a_\alpha \in \mathbb{N},$$

then

$$p(\beta) = \sum_{\alpha \in \Delta} a_\alpha p(\alpha).$$

Consequently, by (10),

$$\beta^* = \sum_{\alpha \in \Delta} a_\alpha \frac{\ell(\alpha)}{\ell(\beta)} \alpha^*,$$

which shows that $\text{ind}(\Delta^*)$ satisfies condition (i') of Remark 3.1.9. □

By Proposition 2.1.1, if α^* is not indivisible, then

$$\alpha^* = 2u^*, \quad u^* \in \text{ind}(\Delta^*) \quad \text{and} \quad u = 2\alpha.$$

It follows that:⁹

Corollary 3.1.11. *Let Δ be a system of simple roots, and let*

$$\beta = \sum_{\alpha \in \Delta} a_\alpha \alpha, \quad a_\alpha \in \mathbb{Z},$$

be the expression of β relative to Δ . Then $\ell(\beta)$ divides $2a_\alpha \ell(\alpha)$; it even divides $a_\alpha \ell(\alpha)$ if α^ is indivisible.*

Before continuing to state the properties of systems of simple roots, we show that such systems exist.

⁹Editors' note: The preceding sentence was added, and in Corollary 3.1.11 the expression $4a_\alpha \ell(\alpha)$ was replaced by $2a_\alpha \ell(\alpha)$.

3.2 Systems of positive roots

Definition 3.2.1. A subset $R_+ \subset R$ is called a *system of positive roots* of $\mathcal{R}$ (or of R ; see Remark 3.2.2) if it satisfies:

$$(P\ 1) \quad R_+ \cap (-R_+) = \emptyset;$$

$$(P\ 2) \quad R_+ \cup (-R_+) = R;$$

$$(P\ 3) \quad (\mathbb{Q}_+ \cdot R_+) \cap R \subset R_+.$$

In particular, such a set is closed. We shall see later, in Corollary 3.3.8, that a closed set satisfying (P 1) and (P 2) also satisfies (P 3), and hence is a system of positive roots. If $w \in W$ and R_+ is a system of positive roots, then $w(R_+)$ is a system of positive roots.

Remark 3.2.2. This definition involves only R . We shall also say that R_+ is a system of positive roots of R .

Remark 3.2.3. Conditions (P 1) and (P 2) immediately give

$$\text{Card}(R_+) = \frac{\text{Card}(R)}{2}.$$

It follows that if R_+ and R'_+ are two systems of positive roots and $R_+ \subset R'_+$, then $R_+ = R'_+$.

Remark 3.2.4. If R_+ is a system of positive roots, then R_+^* is a system of positive coroots (i.e. a system of positive roots of R^*).

This follows immediately from Lemma 1.1.4 and Scholium 1.2.2.

Definition 3.2.5. Let Δ be a system of simple roots. Set

$$\mathcal{P}(\Delta) = (\mathbb{Q}_+ \cdot \Delta) \cap R = (\mathbb{N}_+ \cdot \Delta) \cap R.$$

Proposition 3.2.6. If Δ is a system of simple roots, then $\mathcal{P}(\Delta)$ is a system of positive roots. If Δ is a system of simple roots and R_+ is a system of positive roots, then

$$\Delta \subset R_+ \iff R_+ = \mathcal{P}(\Delta).$$

Proof. The first assertion is immediate. If $\Delta \subset R_+$, then $\mathcal{P}(\Delta) \subset R_+$ by (P 3), and therefore $\mathcal{P}(\Delta) = R_+$ by Remark 3.2.3. The rest is trivial. $\square$

Remark 3.2.7. Systems of positive roots exist: endow $\mathcal{V}(R)$ with the structure of a totally ordered vector space. The set of roots that are ≥ 0 for this order is a system of positive roots.

Theorem 3.2.8. Let R_+ be a system of positive roots. There exists a unique system of simple roots $\mathcal{S}(R_+)$ such that

$$\mathcal{S}(R_+) \subset R_+, \quad \text{i.e.} \quad R_+ = \mathcal{P}(\mathcal{S}(R_+)).$$

Uniqueness follows immediately from:

Lemma 3.2.9. Let Δ be a system of simple roots. An element $\alpha \in \mathcal{P}(\Delta)$ belongs to Δ if and only if α is not the sum of two elements of $\mathcal{P}(\Delta)$.

Proof. This follows immediately from the definitions and Proposition 3.1.2. $\square$

We now prove the existence of $\mathcal{S}(R_+)$. Consider the set of subsets T of R_+ such that

$$T = \text{ind}(T) \quad \text{and} \quad \mathbb{Q}_+ \cdot T \supset R_+.$$

This set is nonempty, since it contains $\text{ind}(R_+)$. Let Δ be a minimal element of this set under inclusion. We show that Δ is a system of simple roots; by Proposition 3.1.5(i), it is enough to show that Δ is a linearly independent subset of $M \otimes \mathbb{Q}$.

Lemma 3.2.10. *If $\alpha, \beta \in \Delta$ and*

$$q\alpha + q'\beta \in R, \quad q, q' \in \mathbb{Q},$$

then $qq' \geq 0$.

Proof. If $qq' < 0$, then, after interchanging α and β if necessary, we may write

$$a\alpha - b\beta \in R_+, \quad a, b \in \mathbb{Q}_+^*.$$

By the defining property of Δ , there is a relation

$$a\alpha - b\beta = \sum_{\gamma \in \Delta} c(\gamma)\gamma, \quad c(\gamma) \in \mathbb{Q}_+.$$

If $a \leq c(\alpha)$, this becomes

$$-\beta = \frac{c(\alpha) - a}{b}\alpha + \sum_{\gamma \neq \alpha} \frac{c(\gamma)}{b}\gamma.$$

Thus $-\beta \in (\mathbb{Q}_+ \cdot \Delta) \cap R$, which is contained in R_+ by (P 3). This gives $\beta \in R_+ \cap (-R_+)$, contradicting (P 1).

If instead $a > c(\alpha)$, then

$$(a - c(\alpha))\alpha = b\beta + \sum_{\gamma \neq \alpha} c(\gamma)\gamma,$$

which shows that

$$\Delta \subset \mathbb{Q}_+ \cdot (\Delta \setminus \{\alpha\}),$$

contrary to the minimality of Δ . □

Recall from Complements 2.3.7 that the bilinear form on $\mathcal{V}(R)$ defined by

$$\langle x, y \rangle = (p(x), y)$$

is a Euclidean inner product. Moreover, for $\alpha \in R$,

$$\langle \alpha, y \rangle = \frac{\ell(\alpha)}{2}(\alpha^*, y),$$

so $\langle \alpha, y \rangle$ and $\langle \alpha^*, y \rangle$ have the same sign.

Lemma 3.2.11. *If $\alpha, \beta \in \Delta$, then*

$$(\beta^*, \alpha) \leq 0 \quad \text{and hence} \quad \langle \beta, \alpha \rangle \leq 0.$$

Proof. Indeed,

$$s_\beta(\alpha) = \alpha - (\beta^*, \alpha)\beta \in R,$$

and therefore $(\beta^*, \alpha) \leq 0$ by Lemma 3.2.10. □

We now show that Δ is linearly independent. Otherwise, Lemma 3.2.11 would imply, exactly as in the proof of Proposition 3.1.5, the existence of a nontrivial relation

$$\sum_{\alpha \in \Delta} m(\alpha)\alpha = 0, \quad m(\alpha) \in \mathbb{N}.$$

If $m(\alpha_0) \geq 1$, then

$$-\alpha_0 = (m(\alpha_0) - 1)\alpha_0 + \sum_{\alpha \neq \alpha_0} m(\alpha)\alpha.$$

By (P 3), this gives $\alpha_0 \in R_+ \cap (-R_+)$, contradicting (P 1). Thus Δ is a base of $\mathcal{R}$, completing the proof of Theorem 3.2.8.

Corollary 3.2.12. *Let R_+ be a system of positive roots, let Δ be a base of $\mathcal{R}$, and let $w \in W$. Then*

$$\begin{aligned} \Delta \subset R_+ &\iff R_+ = \mathcal{P}(\Delta) \iff \Delta = \mathcal{S}(R_+), \\ \mathcal{P}(\text{ind}(\Delta^*)) &= \mathcal{P}(\Delta)^*, \quad \mathcal{S}(R_+^*) = \text{ind}(\mathcal{S}(R_+)^*), \end{aligned}$$

and

$$\mathcal{S}(w(R_+)) = w(\mathcal{S}(R_+)), \quad \mathcal{P}(w(\Delta)) = w(\mathcal{P}(\Delta)).$$

Definition 3.2.13. Once a system of simple roots Δ has been chosen, the elements of $\mathcal{P}(\Delta)$ are called *positive*. Once a system of positive roots R_+ has been chosen, the elements of $\mathcal{S}(R_+)$ are called *simple*.

Corollary 3.2.14. *Let R_+ be a system of positive roots, and let $\alpha \in R_+$. The following conditions are equivalent:*

- (i) α is simple, that is, $\alpha \in \mathcal{S}(R_+)$;
- (ii) α is not the sum of two elements of R_+ ;
- (iii) $R_+ \setminus \{\alpha\}$ is closed.

Proof. The equivalence of (i) and (ii) follows immediately from Lemma 3.2.9; the equivalence of (ii) and (iii) is immediate from Definition 3.1.4. $\square$

Definition 3.2.15. Let Δ be a system of simple roots. The sum of the coefficients in the expression of a root α relative to Δ is called the *order of α relative to Δ* and is denoted $\text{ord}_\Delta(\alpha)$.

One has

$$\alpha \in \mathcal{P}(\Delta) \iff \text{ord}_\Delta(\alpha) > 0, \quad \alpha \in \Delta \iff \text{ord}_\Delta(\alpha) = 1.$$

Lemma 3.2.16. *Let Δ be a system of simple roots and $\alpha \in \mathcal{P}(\Delta)$. There is a sequence $\alpha_i \in \Delta$, $i = 1, \dots, m$, such that*

$$\alpha_1 + \alpha_2 + \dots + \alpha_p \in \mathcal{P}(\Delta) \quad \text{for } p = 1, \dots, m,$$

and

$$\alpha_1 + \alpha_2 + \dots + \alpha_m = \alpha.$$

Moreover, for every sequence satisfying these conditions,

$$m = \text{ord}_\Delta(\alpha).$$

Proof. This is immediate from Proposition 3.1.2. $\square$

3.3 Characterization and conjugacy of systems of positive roots

Lemma 3.3.1. *If $\alpha \in \mathcal{P}(\Delta)$, $\beta \in \Delta$, and α and β are not proportional, then*

$$s_\beta(\alpha) \in \mathcal{P}(\Delta).$$

Proof. One has

$$s_\beta(\alpha) = \alpha - (\beta^*, \alpha)\beta.$$

At least one simple root other than β occurs with a nonzero, and hence strictly positive, coefficient in the expression of α . It occurs with the same coefficient in the expression of $s_\beta(\alpha)$; therefore $s_\beta(\alpha)$ is also positive. $\square$

Corollary 3.3.2. *If $\beta \in \Delta$, then the reflection s_β permutes the elements of $\mathcal{P}(\Delta)$ that are not proportional to β .*

Lemma 3.3.3. *Let*

$$\alpha \in \mathcal{P}(\Delta) \setminus \Delta$$

be indivisible. There exists $\beta \in \Delta$ such that

$$s_\beta(\alpha) \in \mathcal{P}(\Delta) \quad \text{and} \quad \text{ord}_\Delta(s_\beta(\alpha)) < \text{ord}_\Delta(\alpha).$$

Proof. By Lemma 3.1.1, there is a $\beta \in \Delta$ such that $(\beta^*, \alpha) > 0$. Since $\alpha \notin \Delta$ and α is indivisible, it cannot be proportional to β . Thus $s_\beta(\alpha) \in \mathcal{P}(\Delta)$ by Lemma 3.3.1, and

$$\text{ord}_\Delta(s_\beta(\alpha)) = \text{ord}_\Delta(\alpha) - (\beta^*, \alpha) < \text{ord}_\Delta(\alpha).$$

$\square$

Corollary 3.3.4. *If $\alpha \in \mathcal{P}(\Delta)$ is indivisible, then there is a sequence $\beta_p \in \Delta$, $p = 1, \dots, q$, such that*

$$\alpha_p = s_{\beta_p} \cdots s_{\beta_1}(\alpha) \in \mathcal{P}(\Delta) \quad \text{for } p = 1, \dots, q,$$

and $\alpha_q \in \Delta$.

Proof. Apply Lemma 3.3.3 repeatedly, inducting on $\text{ord}_\Delta(\alpha)$. $\square$

Proposition 3.3.5. *The Weyl group is generated by the s_α , for $\alpha \in \Delta$. Every indivisible root is conjugate under an element of the Weyl group to a simple root.*

Proof. The second assertion follows immediately from Corollary 3.3.4. The first then follows from Corollary 1.2.10 and Proposition 2.1.1. $\square$

Proposition 3.3.6. *Let R_+ be a system of positive roots, and let $P' \subset R$ be closed and satisfy (P 2). Then there exists $w \in W$ such that*

$$w(R_+) \subset P'.$$

We first state the corollaries.

Corollary 3.3.7. *The Weyl group acts transitively on the set of systems of positive roots, and on the set of systems of simple roots.*

Corollary 3.3.8. *A subset of R is a system of positive roots if and only if it is closed and satisfies (P 1) and (P 2).*

Corollary 3.3.9. *Suppose that $\Gamma_0(R)$ is endowed with an ordered-group structure for which every root is either > 0 or < 0 . Then the set of positive roots for this order is a system of positive roots.*

Proof of Proposition 3.3.6. Choose $w \in W$ such that $\text{Card}(w(R_+) \cap P')$ is maximal. Replacing R_+ by $w(R_+)$, we may assume that

$$(*) \quad \text{Card}(R_+ \cap P') \geq \text{Card}(R_+ \cap s_\alpha(P')) \quad \text{for every } \alpha \in \mathcal{S}(R_+) = \Delta.$$

We show that $R_+ \subset P'$. Otherwise, since P' is closed, there is an $\alpha \in \Delta$ such that $\alpha \notin P'$. As P' satisfies (P 2), one then has $-\alpha \in P'$, and also $-2\alpha \in P'$ if 2α is a root.

For every $\beta \in R_+ \cap P'$, one has $\beta \neq \alpha$. If 2α is not a root, then $s_\alpha(\beta) \in R_+$ by Lemma 3.3.1; hence

$$s_\alpha(R_+ \cap P') \subset R_+ \cap s_\alpha(P').$$

But $R_+ \cap s_\alpha(P')$ also contains $\alpha = s_\alpha(-\alpha)$, contradicting (*). If 2α is a root, the same argument applies. $\square$

Thus, to study closed sets of roots satisfying (P 2), it is enough to consider those that contain a system of positive roots.

Proposition 3.3.10. *Let R_+ be a system of positive roots, and let P' be a closed subset of R containing R_+ . Put*

$$\Delta = \mathcal{S}(R_+), \quad \Delta' = \Delta \cap (-P').$$

Then P' is the union of R_+ and the set of roots that are linear combinations of the elements of Δ' with negative coefficients.

Proof. We argue by induction on the order of a root $\gamma \in P' \setminus R_+$. If $\text{ord}_\Delta(\gamma) = -1$, then $-\gamma \in \Delta$, and hence $\gamma \in -\Delta'$.

Suppose that $\text{ord}_\Delta(\gamma) < -1$. By Proposition 3.1.2, there exists $\beta \in \Delta$ such that $-\gamma - \beta \in R$. Then

$$0 < \text{ord}_\Delta(-\gamma - \beta) = \text{ord}_\Delta(-\gamma) - 1 < \text{ord}_\Delta(-\gamma).$$

The first inequality shows that $-\gamma - \beta \in R_+$. Consequently, since $R_+ \cap (-R_+) = \emptyset$, and since $\gamma + \beta$ is the sum of two roots of P' , the root $\gamma + \beta$ belongs to $P' \setminus R_+$ and satisfies

$$\text{ord}_\Delta(\gamma + \beta) > \text{ord}_\Delta(\gamma).$$

By induction, $\gamma + \beta$ is a linear combination of the elements of Δ' with negative coefficients. Since $\gamma = (\gamma + \beta) - \beta$, it remains only to show that $\beta \in -P'$. But $\gamma \in P'$ and $-(\gamma + \beta) \in R_+ \subset P'$, so

$$-\beta = \gamma - (\gamma + \beta) \in P'.$$

$\square$

Definition 3.3.10.1. ¹⁰ A subset R' of R is called *symmetric* if

$$-R' = R'.$$

¹⁰Editors' note: This definition was inserted here.

Scholium 3.3.11. *Let P' be a set of roots satisfying (P 2) and closed. There exist a system of simple roots Δ and a subset Δ' of Δ such that*

$$P' = R \cap (\mathbb{N} \cdot \Delta \cup -\mathbb{N} \cdot \Delta').$$

If

$$R_{\Delta'} = (\mathbb{Z} \cdot \Delta') \cap R,$$

which is a closed and symmetric subset of R , then P' is the disjoint union of $R_{\Delta'}$ and the closed subset of $\mathcal{P}(\Delta)$ formed by the

$$\alpha \in \mathcal{P}(\Delta) \setminus \mathbb{N} \cdot \Delta',$$

that is, the positive roots whose decomposition with respect to Δ “contains at least one root of $\Delta \setminus \Delta'$ ”.

3.4 Closed and symmetric sets of roots

Proposition 3.4.1. *Let*

$$\mathcal{R} = (M, M^*, R, R^*)$$

be a root datum, and let R' be a closed and symmetric subset of R . Then:

- (i) $\mathcal{R}' = (M, M^*, R', R'^*)$ is a root datum;
- (ii) for every system of positive roots R_+ of R , $R'_+ = R_+ \cap R'$ is a system of positive roots of R' ;
- (iii) the Weyl group $W(\mathcal{R}')$ of $\mathcal{R}'$ is the subgroup of $W(\mathcal{R})$ generated by the s_α , for $\alpha \in R'$.

Proof. The first assertion is immediate from Corollary 1.2.11; the second follows from Corollary 3.3.8; and the third is evident. $\square$

Corollary 3.4.2. *Let $w \in W(\mathcal{R}')$. The order of w is the smallest integer $n > 0$ such that*

$$w^n(\alpha') = \alpha' \quad \text{for every } \alpha' \in R'.$$

Proof. Apply Corollary 1.2.7 to the root datum $\mathcal{R}'$. $\square$

Corollary 3.4.3. *Let α and β be two nonproportional roots. Let n be the smallest integer > 0 such that*

$$(s_\alpha s_\beta)^n(\alpha) = \alpha, \quad (s_\alpha s_\beta)^n(\beta) = \beta.$$

Then the subgroup $W_{\alpha, \beta}$ of W generated by s_α and s_β is defined by the relations

$$s_\alpha^2 = 1, \quad s_\beta^2 = 1, \quad (s_\alpha s_\beta)^n = 1.$$

Proof. In view of Corollary 3.4.2, it remains only to verify the following lemma. $\square$

Lemma 3.4.4. *Let G be the group generated by two elements x and y subject to the relations $x^2 = y^2 = 1$. Every normal subgroup of G containing neither x nor y is generated, as a normal subgroup, by an element of the form $(xy)^n$, where $n > 0$.*

Proof. ¹¹ Every element of G can be written as $(xy)^n$, as $(yx)^n = (xy)^{-n}$, or in one of the forms

$$\begin{aligned} (a) \quad & x(yx)^{2n} \quad \text{or } y(xy)^{2n+1}, \\ (b) \quad & y(xy)^{2n} \quad \text{or } x(yx)^{2n+1}, \end{aligned}$$

where $n \in \mathbb{N}$. The elements of type (a), respectively of type (b), are conjugate to x , respectively to y . $\square$

Remark 3.4.5. The integer n is computed immediately. Put

$$(\alpha^*, \beta) = p, \quad (\beta^*, \alpha) = q.$$

Then¹²

$$(s_\alpha s_\beta)(\alpha) = (pq - 1)\alpha - q\beta, \quad (s_\alpha s_\beta)(\beta) = p\alpha - \beta.$$

Thus n is the order of the matrix

$$\begin{pmatrix} pq - 1 & p \\ -q & -1 \end{pmatrix}.$$

If $pq = 0$, then $p = q = 0$ by Corollary 2.2.2, whence $n = 2$. Otherwise, Proposition 2.3.1 shows that pq equals 1, 2, or 3, and one finds respectively $n = 3, 4$, or 6.

N.B. Expressing the finiteness of the order of the preceding matrix recovers inequality (13) of Proposition 2.3.1.

Definition 3.4.6. Let Δ be a system of simple roots and let $\Delta' \subset \Delta$. We write

$$R_{\Delta'} = R \cap (\mathbb{Q} \cdot \Delta') = R \cap (\mathbb{Z} \cdot \Delta').$$

Lemma 3.4.7. *The set $R_{\Delta'}$ is closed and symmetric, and Δ' is a system of simple roots of the root datum*

$$(M, M^*, R_{\Delta'}, R_{\Delta'}^*).$$

Its Weyl group is the subgroup $W_{\Delta'}$ of W generated by the s_α , for $\alpha \in \Delta'$. Moreover,

$$\Delta \cap R_{\Delta'} = \Delta'.$$

Proof. Trivial. $\square$

Proposition 3.4.8. *Let $R' \subset R$. The following conditions are equivalent:*

- (i) *There exists a vector subspace V' of V , or of $\mathcal{V}(R)$, such that $R' = R \cap V'$.*
- (ii) *There exist a system of simple roots Δ of R and a subset Δ' of Δ such that $R' = R_{\Delta'}$.*

More precisely, under these conditions, every system of simple roots Δ' of (M, M^, R', R'^*) is contained in a system of simple roots Δ of R , and one has $R' = R_{\Delta'}$.*

Proof. The implication (ii) $\Rightarrow$ (i) is clear. Suppose (i). Then R' is closed and symmetric, so (M, M^*, R', R'^*) is a root datum. Let Δ' be a system of simple roots of this datum, and put

$$R'_+ = \mathcal{P}(\Delta').$$

¹¹Editors' note: The original has been corrected in what follows.

¹²Editors' note: The original has been corrected in what follows.

If $V' = V$, then Δ' is a system of simple roots of $\mathcal{R}$, and there is nothing to prove. Otherwise, there exists $x \in V^*$ such that

$$(x, V') = \{0\}, \quad (x, \alpha) \neq 0 \quad \text{for every } \alpha \in R \setminus R'.$$

Put

$$R_+(x) = \{\alpha \in R \mid (x, \alpha) > 0\}, \quad R_+ = R_+(x) \cup R'_+.$$

For every $\alpha \in R$, one has

$$\begin{aligned} (x, \alpha) > 0 &\iff \alpha \in R_+(x), \\ (x, \alpha) < 0 &\iff \alpha \in -R_+(x), \\ (x, \alpha) = 0 &\iff \alpha \in R'. \end{aligned}$$

It follows at once from Corollary 3.3.8 that R_+ is a system of positive roots of R . Put $\Delta = \mathcal{S}(R_+)$. It is enough to prove that $\Delta' \subset \Delta$. Otherwise, let $\alpha \in \Delta' \setminus \Delta$. By Corollary 3.2.14, there exist $\beta, \gamma \in R_+$ such that

$$\alpha = \beta + \gamma.$$

If $\beta, \gamma \in R_+(x)$, then $\alpha \in R_+(x)$, which is impossible because $(x, \Delta') = 0$. If one of β, γ , say β , belongs to R'_+ , then, since R' is symmetric and closed,

$$\gamma = \alpha - \beta \in R_+ \cap R' = R'_+.$$

But then α is not simple in R'_+ , again a contradiction. $\square$

Lemma 3.4.9. *Under the preceding conditions, let*

$$\alpha \in \mathcal{P}(\Delta) \setminus R'.$$

For every $w \in W_{\Delta'}$, one has

$$w(\alpha) \in \mathcal{P}(\Delta) \setminus R'.$$

Proof. It suffices to verify the assertion for $w = s_\beta$, $\beta \in \Delta'$. It then follows from Lemma 3.3.1 and from $s_\beta(R') = R'$. $\square$

Lemma 3.4.10. *Let $w \in W$. Under the conditions of Proposition 3.4.8, the following conditions are equivalent:*

- (i) $w \in W_{\Delta'}$.
- (ii) For every $m \in M$, one has $w(m) - m \in V'$.
- (iii) For every $\alpha \in R$, one has $w(\alpha) - \alpha \in V'$.

Proof. Clearly (i) implies (ii), and (ii) implies (iii). Suppose (iii). Write

$$w = s_{\alpha_n} \cdots s_{\alpha_1}, \quad \alpha_i \in \Delta,$$

and argue by induction on n that every α_i belongs to Δ' . We may assume that

$$w' = s_{\alpha_{n-1}} \cdots s_{\alpha_1} \in W_{\Delta'}.$$

Then, for every $\alpha \in R$,

$$w(\alpha) - \alpha = w'(\alpha) - \alpha - (\alpha_n^*, w'(\alpha))\alpha_n.$$

Taking $\alpha = w'^{-1}(\alpha_n)$, one obtains $2\alpha_n \in V'$, hence $\alpha_n \in \Delta'$. Therefore

$$w = s_{\alpha_n} w' \in W_{\Delta'}.$$

$\square$

3.5 Miscellaneous remarks

Proposition 3.5.1. *Let R_+ be a system of positive roots. Put*

$$\rho_{R_+} = \frac{1}{2} \sum_{\alpha \in \text{ind}(R_+)} \alpha.$$

Then

$$(\beta^*, \rho_{R_+}) = 1 \quad \text{for every } \beta \in \mathcal{S}(R_+).$$

Proof. One can write

$$2\rho_{R_+} = \beta + \sum_{\substack{\alpha \in \text{ind}(R_+) \\ \alpha \neq \beta}} \alpha.$$

Hence

$$s_\beta(2\rho_{R_+}) = 2\rho_{R_+} - 2\beta$$

by Lemma 3.3.2. □

Corollary 3.5.2. *Put*

$$\rho_{R_+}^* = \frac{1}{2} \sum_{\alpha^* \in \text{ind}(R_+^*)} \alpha^*.$$

Then:

(i) $(\rho_{R_+}^*, \alpha) > 0$ for every $\alpha \in R_+$; equivalently, $\rho_{R_+}^* \in \mathcal{C}(R_+)$, in the notation of Definition 3.6.8.

(ii) For every $\alpha \in R$,

$$\text{ord}_{\mathcal{S}(R_+)}(\alpha) = (\rho_{R_+}^*, \alpha).$$

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Remark 3.5.3. If $w \in W$, then

$$\rho_{w(R_+)} = w(\rho_{R_+}), \quad \rho_{w(R_+)}^* = w(\rho_{R_+}^*).$$

Proposition 3.5.4. *Let α and γ be two nonproportional roots, with α indivisible. There exist a system of simple roots containing α and a root β such that*

$$\gamma = a\alpha + b\beta, \quad a, b \in \mathbb{N}.$$

Proof. Choose a basis of the vector space $\mathcal{V}(R)$ containing $\alpha_1 = \alpha$ and $\alpha_2 = \gamma$, and consider the lexicographic order with respect to this basis. Let R_+ be the set of roots > 0 . It is clear that R_+ is a system of positive roots and¹⁴ that the least element of R_+ not proportional to α is simple. This element has the form

$$\beta = p\alpha + q\gamma, \quad 0 < q \leq 1.$$

Consequently

$$\gamma = q^{-1}\beta - q^{-1}p\alpha.$$

Since $q^{-1} > 0$, one has

$$q^{-1} \in \mathbb{N}^*, \quad -q^{-1}p \in \mathbb{N}.$$

□

¹³Editors' note: By Lemma 3.1.5, it is enough to verify the formula for $\alpha \in \Delta$. Proposition 3.5.1, applied to R^* , then gives $(\rho_{R_+}^*, \alpha) = 1 = \text{ord}_{\mathcal{S}(R_+)}(\alpha)$.

¹⁴Editors' note: The original was modified here to take account of the case in which 2α is a root.

We conclude with two remarks on the group $\Gamma_0(R)$.

Proposition 3.5.5. *Let G be an abelian group, and let $f : R \rightarrow G$ be a map satisfying:*

(i) *if $\alpha \in R$, then $f(-\alpha) = -f(\alpha)$;*

(ii) *if $\alpha, \beta, \alpha + \beta \in R$, then*

$$f(\alpha + \beta) = f(\alpha) + f(\beta).$$

There exists a unique group homomorphism

$$\bar{f} : \Gamma_0(R) \longrightarrow G$$

such that $\bar{f}(\alpha) = f(\alpha)$ for every $\alpha \in R$.

Proof. Let Δ be a system of simple roots of R , and write

$$\beta = \sum_{\alpha \in \Delta} a(\alpha) \alpha.$$

Lemma 3.2.16 gives at once

$$f(\beta) = \sum_{\alpha \in \Delta} a(\alpha) f(\alpha).$$

Now Δ is a basis of $\Gamma_0(R)$. □

Proposition 3.5.6. *Let Δ be a system of simple roots. There exists on $\Gamma_0(R)$ a totally ordered group structure such that the roots > 0 are the elements of $\mathcal{P}(\Delta)$, and such that*

$$\alpha \longmapsto \text{ord}_\Delta(\alpha)$$

is increasing.

Proof. Let

$$\alpha_1, \alpha_2, \dots, \alpha_n, \quad n = \text{rgss}(\mathcal{R}),$$

be the elements of Δ . Every $x \in \Gamma_0(R)$ has a decomposition

$$x = \sum_{i=1}^n m_i(x) \alpha_i.$$

Take the lexicographic order with respect to the functions

$$\sum_i m_i, \quad m_n, \quad m_{n-1}, \quad \dots, \quad m_2.$$

□

Remark 3.5.7. The first roots, in order, are

$$\alpha_1, \alpha_2, \dots, \alpha_n;$$

then come, when they are roots,

$$2\alpha_1, \quad \alpha_1 + \alpha_2, \quad \alpha_1 + \alpha_3, \quad \dots$$

3.6 Weyl chambers

Lemma 3.6.1. *Let V be a finite-dimensional real vector space.¹⁵ Let f_i be independent linear forms. Put*

$$C = \{x \in V \mid f_i(x) > 0\}.$$

Then C is a maximal convex subset of

$$X = V \setminus \bigcup_i f_i^{-1}(0).$$

Proof. Trivial. □

Definition 3.6.2. A subset $C \subset V$ described by the construction of Lemma 3.6.1 is called, here, a *chamber* of V .

Definition 3.6.3. A hyperplane H of V is called a *wall* of C if

$$H \cap (\overline{C} \setminus C)$$

contains a nonempty open subset of H .

Remark 3.6.4. For a convex subset, the closure can be described without reference to the topology of V : it is the set of endpoints of all open segments contained in the given subset.

Lemma 3.6.5. *Under the conditions of Lemma 3.6.1,*

$$\overline{C} = \{x \in V \mid f_i(x) \geq 0\}.$$

The walls of C are the hyperplanes $f_i^{-1}(0)$.

Proof. The first assertion is clear. The second follows from

$$\overline{C} \setminus C \subset \bigcup_i f_i^{-1}(0)$$

and from the fact that the $f_i^{-1}(0)$ are visibly walls of C . □

Proposition 3.6.6. *Let C be a chamber of V . If*

$$H_i, \quad i = 1, 2, \dots, n,$$

are the distinct walls of C , then, for every system of linear forms $\{u_i\}$ such that $H_i = u_i^{-1}(0)$, there exist $\varepsilon_i \in \{-1, +1\}$ for which

$$C = \{x \in V \mid \varepsilon_i u_i(x) > 0\}.$$

For every wall H of C ,

$$H \cap C = \emptyset, \quad H \cap \overline{C} = \overline{C_H},$$

where C_H is a chamber in H . The walls of C_{H_i} are the $H_j \cap H_i$, for $j \neq i$.

Proof. This follows immediately from the lemma. □

Definition 3.6.7. The C_{H_i} are called the *faces* of C .

¹⁵Editors' note: The original $\mathbb{Q}$ has been replaced by $\mathbb{R}$.

Let now

$$\mathcal{R} = (M, M^*, R, R^*)$$

be a root datum, and put

$$V_{\mathbb{R}}^* = M^* \otimes \mathbb{R}.$$

Definition 3.6.8. ¹⁶ For every $\alpha \in R$, put

$$\mathcal{H}_{\alpha} = \{x \in V_{\mathbb{R}}^* \mid (x, \alpha) = 0\}.$$

Write

$$X = V_{\mathbb{R}}^* \setminus \bigcup_{\alpha \in R} \mathcal{H}_{\alpha}.$$

For every $x \in X$, put

$$R_+(x) = \{\alpha \in R \mid (x, \alpha) > 0\}.$$

For every system of positive roots R_+ , put

$$\mathcal{C}(R_+) = \{x \in V_{\mathbb{R}}^* \mid (x, \alpha) > 0 \text{ for every } \alpha \in R_+\}.$$

Proposition 3.6.9. (i) For every $x \in X$, $R_+(x)$ is a system of positive roots. For every system of positive roots R_+ , $\mathcal{C}(R_+)$ is a chamber in $V_{\mathbb{R}}^*$. The $\mathcal{C}(R_+)$ are the maximal convex subsets of X .

(ii) Let Δ be a system of simple roots. One has

$$\mathcal{C}(\mathcal{P}(\Delta)) = \{x \in V_{\mathbb{R}}^* \mid (x, \alpha) > 0 \text{ for every } \alpha \in \Delta\}.$$

The walls of $\mathcal{C}(\mathcal{P}(\Delta))$ are the hyperplanes

$$\mathcal{H}_{\alpha} = \alpha^{-1}(0), \quad \alpha \in \Delta;$$

its faces are

$$C_{\alpha} = \{x \in V_{\mathbb{R}}^* \mid (x, \alpha) = 0, (x, \beta) > 0 \text{ for } \beta \in \Delta, \beta \neq \alpha\}.$$

(iii) One has the equivalence

$$R_+(x) = R_+ \iff x \in \mathcal{C}(R_+).$$

Proof. It is first clear that $R_+(x)$ is a system of positive roots. Since $x \in \mathcal{C}(R_+(x))$, the union of the $\mathcal{C}(R_+(x))$ is X . Property (iii) is immediate; it follows that the $\mathcal{C}(R_+)$ form a partition of X . The first assertion of (ii) is evident. It follows at once that $\mathcal{C}(R_+)$ is a chamber in $V_{\mathbb{R}}^*$, which proves the remainder of (i). Put

$$C = \mathcal{C}(\mathcal{P}(\Delta)).$$

It remains only to note that

$$\overline{C} \cap \left(V_{\mathbb{R}}^* \setminus \bigcup_{\alpha \in R} \alpha^{-1}(0) \right) = \overline{C} \cap \left(V_{\mathbb{R}}^* \setminus \bigcup_{\alpha \in \Delta} \alpha^{-1}(0) \right)$$

to complete the proof of (ii) by Lemma 3.6.1. □

¹⁶Editors' note: The definition of the hyperplanes $\mathcal{H}_{\alpha}$ was added here.

Definition 3.6.10. The $\mathcal{C}(R_+)$ are called the *Weyl chambers* of the root datum. Conversely, for every Weyl chamber C , put

$$R_+(C) = R_+(x)$$

for any $x \in C$.

Corollary 3.6.11. *The maps*

$$R_+ \longmapsto \mathcal{C}(R_+), \quad C \longmapsto R_+(C)$$

give a bijective correspondence between systems of positive roots and Weyl chambers.

This correspondence is invariant under the Weyl group:

Lemma 3.6.12. *If $w \in W(\mathcal{R})$, then*

$$\mathcal{C}(w(R_+)) = w(\mathcal{C}(R_+)).$$

Corollary 3.6.13. *The correspondences*

$$\Delta \longleftrightarrow R_+ \longleftrightarrow C$$

are isomorphisms of homogeneous spaces under $W(\mathcal{R})$.

We shall see later that these homogeneous spaces are principal (5.5).

Remark 3.6.14. If C is a Weyl chamber, then $-C$ is also one; it is called the *opposite* of C . Consequently there exists $w_0 \in W$ —and in fact exactly one, cf. 5.5—such that

$$w_0(C) = -C.$$

It is called the *symmetry* of the root datum relative to the Weyl chamber C (or relative to $R_+(C)$, or to $\mathcal{S}(R_+(C))$, and so on).

4 Reduced root data of semisimple rank 2

4.0.¹⁷ Let $\mathcal{R}$ be a root datum of semisimple rank 2. Let $\{\alpha, \beta\}$ be a system of simple roots. Suppose that $\ell(\alpha) \leq \ell(\beta)$. By 2.3.1 and 3.2.1, there are then four possibilities:

Type	$\ell(\beta)/\ell(\alpha)$	$\ell(\beta^*)/\ell(\alpha^*)$	(β^*, α)	(α^*, β)
$A_1 \times A_1$	—	—	0	0
A_2	1	1	−1	−1
B_2	2	1/2	−1	−2
G_2	3	1/3	−1	−3

It follows from Remark 3.4.5 that the order of $s_\alpha s_\beta$ is respectively 2, 3, 4, 6.

We study each of these systems separately and give the list of indivisible roots.

Type $A_1 \times A_1$. The indivisible roots are

$$\alpha, \quad \beta, \quad -\alpha, \quad -\beta.$$

¹⁷Editors' note: The numbering 4.0 was added for later references.

The corresponding coroots are

$$\alpha^*, \quad \beta^*, \quad -\alpha^*, \quad -\beta^*.$$

Type A_2 . The indivisible positive roots are as follows:

root γ	α	β	$\alpha + \beta$
$\ell(\gamma)/\ell(\alpha)$	1	1	1
coroot γ^*	α^*	β^*	$\alpha^* + \beta^*$
$\ell(\gamma^*)/\ell(\beta^*)$	1	1	1

The half-sum of the indivisible positive roots is

$$\rho = \alpha + \beta.$$

Type B_2 . The indivisible positive roots are as follows:

root γ	α	β	$\alpha + \beta$	$2\alpha + \beta$
$\ell(\gamma)/\ell(\alpha)$	1	2	1	2
coroot γ^*	α^*	β^*	$2\alpha^* + \beta^*$	$\alpha^* + \beta^*$
$\ell(\gamma^*)/\ell(\beta^*)$	2	1	2	1

The half-sum of the indivisible positive roots is

$$\rho = \frac{4\alpha + 3\beta}{2}.$$

Type G_2 . The indivisible positive roots are as follows:

root γ	α	β	$\alpha + \beta$	$2\alpha + \beta$	$3\alpha + \beta$	$3\alpha + 2\beta$
$\ell(\gamma)/\ell(\alpha)$	1	3	1	1	3	3
coroot γ^*	α^*	β^*	$\alpha^* + 3\beta^*$	$2\alpha^* + 3\beta^*$	$\alpha^* + \beta^*$	$\alpha^* + 2\beta^*$
$\ell(\gamma^*)/\ell(\beta^*)$	3	1	3	3	1	1

The half-sum of the indivisible positive roots is

$$\rho = 5\alpha + 3\beta.$$

Proposition 4.1. ¹⁸ Let n be the order of $s_\alpha s_\beta$. Put $u_0 = 0$, and, for $p \in \mathbb{N}$, define

$$\begin{cases} u_{2p+1} = u_{2p} + (s_\alpha s_\beta)^p(\alpha), \\ u_{2p+2} = u_{2p+1} + (s_\alpha s_\beta)^p s_\alpha(\beta). \end{cases}$$

Then: ¹⁹

¹⁸The following numbers 4.1 to 4.4 are used in the proof of Theorem 5.1. There are now simpler proofs of that theorem; see *Groupes et algèbres de Lie*, Chapter V, §3, Theorem 1. Editors' note: The reference was made explicit and this note, which occurred at 4.2 in the original, was moved here.

¹⁹Editors' note: Let ρ be the half-sum of the positive roots, cf. Proposition 3.5.1. If, as below, one puts $w_0 = 1$, $w_1 = s_\alpha$, $w_2 = s_\beta s_\alpha$, $w_3 = s_\alpha s_\beta s_\alpha$, and so on, then $u_k = \rho - w_k^{-1}(\rho)$. This proves (i), since $w_{2n} = 1$.

- (i) $u_{k+2n} = u_k$, for every $k \in \mathbb{N}$.
- (ii) $u_0 = 0$, $u_1 = \alpha$, $u_{2n-1} = \beta$, and $u_{2n} = 0$.
- (iii) If $1 < k < 2n - 1$, then

$$u_k = a_k \alpha + b_k \beta, \quad a_k, b_k \in \mathbb{N}^*.$$

Proof. Assertion (i) follows from $(s_\alpha s_\beta)^n = 1$ and $u_{2n} = 0$.

We prove (ii) and (iii). Direct calculation gives, in the four cases, the following sequences of values: ²⁰

$$\begin{array}{l|l} A_1 \times A_1 & 0, \alpha, \beta + \alpha, \beta, 0, \\ A_2 & 0, \alpha, \beta + 2\alpha, 2\beta + 2\alpha, 2\beta + \alpha, \beta, 0, \\ B_2 & 0, \alpha, \beta + 3\alpha, 2\beta + 4\alpha, 3\beta + 4\alpha, 3\beta + 3\alpha, 2\beta + \alpha, \beta, 0, \\ G_2 & 0, \alpha, \beta + 4\alpha, 2\beta + 6\alpha, 4\beta + 9\alpha, 5\beta + 10\alpha, 6\beta + 10\alpha, \\ & 6\beta + 9\alpha, 5\beta + 6\alpha, 4\beta + 4\alpha, 2\beta + \alpha, \beta, 0. \end{array}$$

□

Lemma 4.2. *Put*

$$w_{2p} = (s_\beta s_\alpha)^p, \quad w_{2p+1} = s_\alpha (s_\beta s_\alpha)^p,$$

so that $w_0, \dots, w_{2n-1}$ are the distinct elements of W . Let $u_0, \dots, u_{2n-1}$ be the elements of V defined in Proposition 4.1. For every $x \in V^*$, one has

$$x - w_k(x) = n_k \alpha^* + m_k \beta^*,$$

where $n_k, m_k \in \mathbb{Q}$ and

$$n_k + m_k = (x, u_k).$$

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Proof. The proof is by induction on k . If $k = 0$, the formula is trivial. We carry out, for example, the passage from w_{2p} to w_{2p+1} . Since $w_{2p+1} = s_\alpha w_{2p}$,

$$\begin{aligned} x - w_{2p+1}(x) &= x - w_{2p}(x) + w_{2p}(x) - s_\alpha w_{2p}(x) \\ &= n_{2p} \alpha^* + m_{2p} \beta^* + (w_{2p}(x), \alpha) \alpha^*. \end{aligned}$$

Consequently

$$\begin{aligned} n_{2p+1} + m_{2p+1} &= n_{2p} + m_{2p} + ((s_\beta s_\alpha)^p(x), \alpha) \\ &= (x, u_{2p}) + (x, (s_\alpha s_\beta)^p(\alpha)) \\ &= (x, u_{2p+1}). \end{aligned}$$

□

²⁰Editors' note: The original was corrected here to agree with the convention $\ell(\alpha) \leq \ell(\beta)$ of 4.0.

²¹Editors' note: In view of the preceding editor's note, this follows immediately from Propositions 3.5.1 and 1.1.9: one has

$$n_k + m_k = (x - w_k(x), \rho) = (x, \rho - w_k^{-1}(\rho)) = (x, u_k).$$

Corollary 4.3. ²² Let $x \in V^*$. For every $w \in W$, put

$$x - w(x) = a_w \alpha^* + b_w \beta^*.$$

If $(x, \alpha) \geq 0$ and $(x, \beta) \geq 0$, then

$$a_w + b_w \geq 0.$$

If moreover $(x, \alpha) > 0$ (respectively $(x, \beta) > 0$), then

$$a_w + b_w > 0$$

for $w \neq 1, s_\beta$ (respectively for $w \neq 1, s_\alpha$).

Proof. This follows at once from Proposition 4.1 and Lemma 4.2. $\square$

Corollary 4.4. Let $\mathcal{R}$ be any root datum, and let Δ be a system of simple roots. Let γ be a positive root, let α, β be two simple roots, and let $W_{\alpha, \beta}$ be the subgroup of W generated by s_α and s_β . If

$$\text{ord}_\Delta(s_\alpha(\gamma)) < \text{ord}_\Delta(\gamma), \quad \text{ord}_\Delta(s_\beta(\gamma)) \leq \text{ord}_\Delta(\gamma),$$

then, for every $w \in W_{\alpha, \beta}$,

$$\text{ord}_\Delta(w(\gamma)) \leq \text{ord}_\Delta(\gamma).$$

Moreover,

$$\text{ord}_\Delta(w(\gamma)) < \text{ord}_\Delta(\gamma)$$

if $w \neq 1, s_\beta$.

Proof. Consider the dual root datum $\mathcal{R}^*$, and then the datum

$$(M^*, M, R'^*, R'),$$

where R' is the set of roots that are rational linear combinations of α and β . Applying Corollary 4.3 to this datum gives the stated result. ²³ $\square$

5 The Weyl group: generators and relations

Let $\mathcal{R}$ be a root datum. Since the Weyl group is the same for this datum and for the corresponding reduced datum, we may assume $\mathcal{R}$ reduced when studying the Weyl group.

Let

$$\Delta = \{\alpha_1, \dots, \alpha_n\}$$

be a system of simple roots, where $n = \text{rgss}(\mathcal{R})$. Let n_{ij} be the order of $s_{\alpha_i} s_{\alpha_j}$ in W . In particular $n_{ii} = 1$, and Remarks 3.4.2 and 3.4.3 show that the subgroup W_{ij} of W generated by s_{α_i} and s_{α_j} is defined by

$$s_{\alpha_i}^2 = s_{\alpha_j}^2 = (s_{\alpha_i} s_{\alpha_j})^{n_{ij}} = 1.$$

Theorem 5.1. The group W is the group generated by the elements

$$s_{\alpha_i}, \quad i = 1, 2, \dots, n,$$

subject to the relations

$$(s_{\alpha_i} s_{\alpha_j})^{n_{ij}} = 1.$$

²²Editors' note: The statement was corrected.

²³Editors' note: The hypotheses say that $(\gamma, \alpha^*) > 0$ and $(\gamma, \beta^*) \geq 0$. Consequently $\gamma - w(\gamma) \in \mathbb{N}\alpha + \mathbb{N}\beta$ for every $w \in W_{\alpha, \beta}$, and it is nonzero if $w \notin \{1, s_\beta\}$.

Proof. We have already seen that the theorem is true when $n = 2$; we shall use this fact in the proof. Introduce the group $\widetilde{W}$ generated by elements

$$T_i, \quad i = 1, 2, \dots, n,$$

subject to the relations

$$(T_i T_j)^{n_{ij}} = 1.$$

In particular $n_{ii} = 1$, whence $T_i^2 = 1$. Let

$$p : \widetilde{W} \longrightarrow W$$

be the group homomorphism sending T_i to s_{α_i} . We know that p is surjective; we shall prove that it is injective. $\square$

Lemma 5.2. *For every $\alpha \in \mathcal{P}(\Delta)$, one can define uniquely an element $T_\alpha \in \widetilde{W}$ with the following properties:*

- (i) $p(T_\alpha) = s_\alpha$;
- (ii) $T_{\alpha_i} = T_i$;
- (iii) if β and α are positive roots such that $s_{\alpha_i}(\alpha) = \beta$, then

$$T_i T_\alpha T_i = T_\beta.$$

Proof. First observe that it follows from 1.2.10 and 3.3.6 that (i) is a consequence of (ii) and (iii), and that (ii) and (iii) determine the T_α completely. We construct them by induction on $\text{ord}_\Delta(\alpha)$. If $\text{ord}_\Delta(\alpha) = 1$, then $\alpha \in \Delta$, and we put $T_\alpha = T_i$ if $\alpha = \alpha_i$.

Consider the hypothesis

- (H_p) there exist T_α , for $\alpha \in \mathcal{P}(\Delta)$ with $\text{ord}_\Delta(\alpha) \leq p$, satisfying (ii), and satisfying (iii) whenever $\text{ord}_\Delta(\alpha) \leq p$ and $\text{ord}_\Delta(\beta) \leq p$.

It holds for $p = 1$. Indeed, if α and $s_{\alpha_i}(\alpha) = \beta$ are simple, then α_i and α are orthogonal. Thus, if $\alpha_j = \alpha = \beta$, one has $n_{ij} = 2$, whence

$$T_i T_j T_i = T_j.$$

Suppose $p > 1$ and (H_{p-1}) holds.

A) Construction of the T_α for $\text{ord}_\Delta(\alpha) \leq p$. It is enough to do this when $\text{ord}_\Delta(\alpha) = p$. There exists $\alpha_i \in \Delta$ such that

$$s_{\alpha_i}(\alpha) \in \mathcal{P}(\Delta), \quad \text{ord}_\Delta(s_{\alpha_i}(\alpha)) < p$$

by Lemma 3.3.3. Put

$$(\star) \quad T_\alpha = T_i T_{s_{\alpha_i}(\alpha)} T_i.$$

We verify that T_α depends only on α . Let $\alpha_j \in \Delta$ also satisfy

$$s_{\alpha_j}(\alpha) \in \mathcal{P}(\Delta), \quad \text{ord}_\Delta(s_{\alpha_j}(\alpha)) < p.$$

We must prove

$$(\dagger) \quad T_i T_{s_{\alpha_i}(\alpha)} T_i = T_j T_{s_{\alpha_j}(\alpha)} T_j.$$

We distinguish two cases.

- (1) Suppose that α is a linear combination of α_i and α_j . The same is then true of $s_{\alpha_i}(\alpha)$ and $s_{\alpha_j}(\alpha)$; by (H_{p-1}) , the elements $T_{s_{\alpha_i}(\alpha)}$ and $T_{s_{\alpha_j}(\alpha)}$ are words in T_i, T_j . Since the projection of $(+)$ to $\widetilde{W}$ holds and the theorem is true for $n = 2$, the restriction of p to the subgroup of $\widetilde{W}$ generated by T_i, T_j is injective. Thus $(+)$ holds.
- (2) Suppose that α is not a linear combination of α_i and α_j . Then, for $w \in W_{\alpha_i, \alpha_j}$, all the $w(\alpha)$ are positive (cf. Lemma 3.4.9). The relation to be verified can also be written

$$(++) \quad (T_i T_j)^{n_{ij}-1} T_{s_{\alpha_i}(\alpha)} (T_j T_i)^{n_{ij}-1} = T_{s_{\alpha_j}(\alpha)}.$$

Corollary 4.4 shows that all the $w(\alpha)$ have order $< p$ for $w \in W_{\alpha_i, \alpha_j}$, $w \neq 1$. We may therefore apply (H_{p-1}) $2(n_{ij} - 1)$ times, which proves the relation.

B) Verification of (H_p) . ²⁴ We must verify that, if $\alpha_j \in \Delta$ and $\beta = s_{\alpha_j}(\alpha)$ satisfies $\text{ord}_\Delta(\beta) \leq p$, then

$$T_j T_\alpha T_j = T_\beta.$$

If $\text{ord}_\Delta(\beta) < p$, this follows from the preceding construction, since $T_j^2 = 1$. We may therefore suppose

$$\text{ord}_\Delta(\beta) = p = \text{ord}_\Delta(\alpha).$$

In this case α and α_j are orthogonal, so $\beta = \alpha$, and it remains to prove

$$(\dagger) \quad T_j T_\alpha T_j = T_\alpha.$$

By $(\star)$,

$$T_\alpha = T_i T_{s_{\alpha_i}(\alpha)} T_i,$$

so it remains to verify

$$(+++) \quad T_j T_i T_{s_{\alpha_i}(\alpha)} T_i T_j = T_\alpha = T_i T_{s_{\alpha_i}(\alpha)} T_i.$$

□

Proof of Lemma 5.2, continued. Put $m = n_{ij}$, $s_i = s_{\alpha_i}$, and $s_j = s_{\alpha_j}$. We have

$$T_j T_i = (T_i T_j)^{m-1}.$$

By Corollary 4.4, $\text{ord}_\Delta(w(\alpha)) < p$ for every $w \in W_{ij}$ distinct from

$$1 = (s_i s_j)^m \quad \text{and} \quad s_j = s_i (s_i s_j)^{m-1}.$$

The induction hypothesis therefore gives

$$\begin{aligned} T_j (T_i T_j)^{m-2} T_{s_i(\alpha)} (T_j T_i)^{m-2} T_j \\ = T_{s_j(s_i s_j)^{m-2} s_i(\alpha)} = T_{s_i s_j(\alpha)} = T_{s_i(\alpha)}. \end{aligned}$$

The last equality holds because $s_j(\alpha) = \alpha$. Hence

$$(T_i T_j)^{m-1} T_{s_i(\alpha)} (T_j T_i)^{m-1} = T_i T_{s_i(\alpha)} T_i,$$

which proves $(+++)$. □

²⁴Editors' note: The argument in the original has been expanded in what follows.

Lemma 5.3. *Let $h \in \widetilde{W}$. Write*

$$h = T_{\alpha_1} \cdots T_{\alpha_m},$$

where the $\alpha_i \in \Delta$, not necessarily distinct, are chosen so that m is minimal. Then

$$p(h)(\alpha_m) \in -\mathcal{P}(\Delta).$$

Proof. Since

$$p(T_{\alpha_m})(\alpha_m) = s_{\alpha_m}(\alpha_m) = -\alpha_m,$$

if $p(h)(\alpha_m)$ were positive, there would be an index k , $1 \leq k \leq m-1$, such that

$$u = s_{\alpha_{k+1}} \cdots s_{\alpha_m}(\alpha_m) = -s_{\alpha_{k+1}} \cdots s_{\alpha_{m-1}}(\alpha_m) \in -\mathcal{P}(\Delta)$$

and $s_{\alpha_k}(u) \in \mathcal{P}(\Delta)$. Necessarily $u = -\alpha_k$, by Lemma 3.3.1, whence

$$s_{\alpha_{k+1}} \cdots s_{\alpha_{m-1}}(\alpha_m) = \alpha_k.$$

Property (iii) of Lemma 5.2 then gives

$$T_{\alpha_k} T_{\alpha_{k+1}} \cdots T_{\alpha_{m-1}} T_{\alpha_m} = T_{\alpha_{k+1}} \cdots T_{\alpha_{m-1}},$$

contradicting the minimality of m . □

Let now $h \in \widetilde{W}$ satisfy

$$p(h)(\mathcal{P}(\Delta)) \subset \mathcal{P}(\Delta).$$

Lemma 5.3 gives $p(h) = 1$. This proves Theorem 5.1 and also the following two corollaries.

Corollary 5.4. *If R_+ is a system of positive roots and $w \in W$ satisfies $w(R_+) = R_+$, then $w = 1$.*

Corollary 5.5. *The Weyl group acts freely and transitively on the set of systems of positive roots, respectively on the set of systems of simple roots, respectively on the set of Weyl chambers.*

Choose a system of simple roots Δ , and put

$$R^+ = \mathcal{P}(\Delta).$$

²⁵ For every pair of simple roots $(\alpha, \beta) \in \Delta \times \Delta$, let $R_{\alpha, \beta}$ be the set of roots that are linear combinations of α and β . Put

$$R_{\alpha, \beta}^+ = R^+ \cap R_{\alpha, \beta},$$

and let $W_{\alpha, \beta}$ be the Weyl group of $R_{\alpha, \beta}$, that is, the subgroup of W generated by s_α and s_β .

Theorem 5.6 (Tits). *Let α and β be two simple roots, and let $w \in W$ satisfy $w(\alpha) = \beta$. There exist a sequence of simple roots $\alpha_0, \dots, \alpha_m$ and a sequence $w_0, \dots, w_{m-1}$ of elements of W satisfying:*

- (i) $\alpha_0 = \alpha$ and $\alpha_m = \beta$;
- (ii) $w = w_{m-1} w_{m-2} \cdots w_0$;

²⁵Editors' note: The superscript notation R^+ , rather than R_+ , is used here so that $R_{\alpha, \beta}^+ = R_{\alpha, \beta} \cap R^+$.

- (iii) $w_i(\alpha_i) = \alpha_{i+1}$ for $0 \leq i \leq m-1$;
- (iv) for every i , $0 \leq i \leq m-1$, such that $\alpha_i \neq \alpha_{i+1}$, one has $w_i \in W_{\alpha_i, \alpha_{i+1}}$;
- (v) for every i , $0 \leq i \leq m-1$, such that $\alpha_i = \alpha_{i+1}$, there exists a simple root β_i such that $w_i \in W_{\alpha_i, \beta_i}$.

Proof. Put²⁶

$$M(w) = \text{Card}(R^+ \cap w^{-1}(-R^+)) = \text{Card}\{\gamma \in R^+ \mid w(\gamma) \in -R^+\}.$$

If $M(w) = 0$, then $w(R^+) = R^+$, so $w = 1$ by Corollary 5.4; the theorem is then trivial: $m = 0$, and assertions (iii)–(v) are empty.

We argue by induction on $M(w)$. Suppose $M(w) > 0$. There exists $\beta_0 \in \Delta$ such that $w(\beta_0) \in -R^+$. Put $\alpha_0 = \alpha$, and consider

$$A = w^{-1}(R^+) \cap R_{\alpha_0, \beta_0}.$$

This is a system of positive roots of R_{α_0, β_0} . Hence there exists $w_0 \in W_{\alpha_0, \beta_0}$ such that

$$w_0^{-1}(R_{\alpha_0, \beta_0}^+) = A.$$

Put $w' = ww_0^{-1}$. Lemma 3.4.9 gives

$$R^+ - R_{\alpha_0, \beta_0}^+ = w_0(R^+ - R_{\alpha_0, \beta_0}^+),$$

and therefore

$$\begin{aligned} (1) \quad & (R^+ - R_{\alpha_0, \beta_0}^+) \cap w^{-1}(-R^+) \\ &= w_0\left((R^+ - R_{\alpha_0, \beta_0}^+) \cap w'^{-1}(-R^+)\right). \end{aligned}$$

On the other hand,

$$(2) \quad \beta_0 \in R_{\alpha_0, \beta_0}^+ \cap w^{-1}(-R^+).$$

Since $w_0(R_{\alpha_0, \beta_0}) = R_{\alpha_0, \beta_0}$, one has

$$w_0(-A) = R_{\alpha_0, \beta_0} \cap w'^{-1}(-R^+),$$

and hence

$$\begin{aligned} (2') \quad & R_{\alpha_0, \beta_0}^+ \cap w'^{-1}(-R^+) = R_{\alpha_0, \beta_0}^+ \cap w_0(-A) \\ &= R_{\alpha_0, \beta_0}^+ \cap (-R_{\alpha_0, \beta_0}^+) = \emptyset. \end{aligned}$$

It follows from (1), (2), and (2') that

$$M(w') < M(w).$$

Put $\alpha_1 = w_0(\alpha_0)$. We show that $\alpha_1 \in \Delta$, or equivalently that $\alpha_0 \in w_0^{-1}(\Delta)$. Since $w(\alpha_0) \in \Delta$, we have

$$\alpha_0 \in w^{-1}(\Delta) \cap R_{\alpha_0, \beta_0}.$$

²⁶Editors' note: In what follows, the original argument has been expanded, and equality (1) has been corrected.

Thus α_0 is a simple root of

$$A = w^{-1}(R^+) \cap R_{\alpha_0, \beta_0} = w_0^{-1}(R_{\alpha_0, \beta_0}^+),$$

and consequently belongs to

$$w_0^{-1}(\Delta \cap R_{\alpha_0, \beta_0}^+) = w_0^{-1}\{\alpha_0, \beta_0\}$$

by Lemma 3.4.8. Hence $\alpha_1 = w_0(\alpha_0)$ equals α_0 or β_0 . If $\alpha_1 \neq \alpha_0$, then $\alpha_1 = \beta_0$ and $w_0 \in W_{\alpha_0, \alpha_1}$.

Finally,

$$\beta = w'(\alpha_1) \quad \text{and} \quad M(w') < M(w),$$

so the induction hypothesis completes the proof. $\square$

6 Morphisms of root data

6.1 Definition

Let

$$\mathcal{R} = (M, M^*, R, R^*), \quad \mathcal{R}' = (M', M'^*, R', R'^*)$$

be two root data. Let $f : M' \rightarrow M$ be a linear map and ${}^t f : M^* \rightarrow M'^*$ its transpose.

Definition 6.1.1. The map f is called a *morphism* from $\mathcal{R}'$ to $\mathcal{R}$, and is denoted

$$f : \mathcal{R}' \longrightarrow \mathcal{R},$$

if f induces a bijection from R' onto R , and ${}^t f$ induces a bijection from R^* onto R'^* .

Then ${}^t f$ is a morphism of the dual root data:

$${}^t f : \mathcal{R}^* \longrightarrow \mathcal{R}'^*.$$

If $f : \mathcal{R}' \rightarrow \mathcal{R}$ is a morphism and $\alpha = f(\alpha')$ for $\alpha' \in R'$, then

$$\alpha'^* = {}^t f(\alpha^*).$$

Indeed, if p and p' denote the maps of 1.2.1 for $\mathcal{R}$ and $\mathcal{R}'$, respectively, then

$$p' = {}^t f \circ p \circ f,$$

which gives the assertion at once. The reader may verify the statements below, almost all of which are immediate.

Proposition 6.1.2. *Let $f : \mathcal{R}' \rightarrow \mathcal{R}$ be a morphism of root data. If $\alpha' \in R'$ and $\alpha = f(\alpha')$, then*

$$\alpha'^* = {}^t f(\alpha^*).$$

Moreover, f induces isomorphisms

$$R' \xrightarrow{\sim} R, \quad \Gamma_0(R') \xrightarrow{\sim} \Gamma_0(R), \quad \mathcal{V}(R') \xrightarrow{\sim} \mathcal{V}(R),$$

and ${}^t f$ induces isomorphisms

$$R^* \xrightarrow{\sim} R'^*, \quad \Gamma_0(R^*) \xrightarrow{\sim} \Gamma_0(R'^*), \quad \mathcal{V}(R^*) \xrightarrow{\sim} \mathcal{V}(R'^*).$$

The last is the transpose of the corresponding morphism induced by f . The map

$$s_{\alpha'} \longmapsto s_{f(\alpha')}$$

extends to an isomorphism

$$W(\mathcal{R}') \xrightarrow{\sim} W(\mathcal{R})$$

compatible with the actions of these two groups on the sets of 1.1.13.

Proposition 6.1.3. *The maps*

$$\Delta' \longmapsto f(\Delta'), \quad R'_+ \longmapsto f(R'_+), \quad C' \longmapsto ({}^t f \otimes \mathbb{R})^{-1}(C')$$

define bijective correspondences between the systems of simple roots, the systems of positive roots, and the Weyl chambers for $\mathcal{R}'$ and $\mathcal{R}$. These correspondences are compatible with the actions of the Weyl groups and with the correspondences

$$\mathcal{S}(R_+) \longleftrightarrow R_+ \longleftrightarrow \mathcal{C}(R_+).$$

Lemma 6.1.4. *Morphisms compose. A morphism $f : \mathcal{R}' \rightarrow \mathcal{R}$ is an isomorphism if and only if the linear map $f : M' \rightarrow M$ is bijective.*

6.2 Isogenies

Definition 6.2.1. A morphism $f : \mathcal{R}' \rightarrow \mathcal{R}$ of root data is called an *isogeny* if $f : M' \rightarrow M$ is injective with finite cokernel.

If f is an isogeny, then ${}^t f$ is an isogeny.

Definition 6.2.2. Let $f : \mathcal{R}' \rightarrow \mathcal{R}$ be an isogeny. Put

$$K(f) = \text{Coker}(M' \xrightarrow{f} M).$$

Lemma 6.2.3. *There is a natural pairing*

$$K(f) \times K({}^t f) \longrightarrow \mathbb{Q}/\mathbb{Z}$$

that puts these two finite groups in duality.

Proof. This is classical. □

Lemma 6.2.4. *If $f : \mathcal{R}' \rightarrow \mathcal{R}$ is a morphism, then*

$$\text{rgss}(\mathcal{R}') = \text{rgss}(\mathcal{R}).$$

If f is moreover an isogeny, then

$$\text{rgred}(\mathcal{R}') = \text{rgred}(\mathcal{R}).$$

Proof. This is immediate. □

Lemma 6.2.5. *Every morphism of semisimple root data is an isogeny.*

Proof. The map f must induce an isomorphism from $V' = \mathcal{V}(R')$ onto $V = \mathcal{V}(R)$. □

If $\mathcal{R}'$ and $\mathcal{R}$ are semisimple, every isogeny $f : \mathcal{R}' \rightarrow \mathcal{R}$ defines a commutative diagram:

$$\begin{array}{ccc}
\Gamma_0(R') & \xrightarrow{\sim} & \Gamma_0(R) \\
\downarrow & & \downarrow \\
M' & \xrightarrow{f} & M
\end{array}$$

If $M = \Gamma_0(R)$, then f is necessarily an isomorphism.

Definition 6.2.6. A root datum is called *adjoint*, respectively *simply connected*, if

$$M = \Gamma_0(R), \quad \text{respectively} \quad M^* = \Gamma_0(R^*).$$

An adjoint or simply connected root datum is therefore semisimple. Moreover, $\mathcal{R}$ is adjoint, respectively simply connected, if and only if $\mathcal{R}^*$ is simply connected, respectively adjoint. The preceding result gives the following proposition.

Proposition 6.2.7. *Let $\mathcal{R}$ be a semisimple root datum. The following conditions are equivalent:*

- (i) $\mathcal{R}$ is adjoint, respectively simply connected;
- (ii) every isogeny $\mathcal{R}' \rightarrow \mathcal{R}$, respectively $\mathcal{R} \rightarrow \mathcal{R}'$, is an isomorphism.

Proposition 6.2.8. *Let $\mathcal{R}$ be an adjoint, respectively simply connected, root datum. Every indivisible root, respectively coroot, is an indivisible element of M , respectively of M^* .*

Proof. Every indivisible root belongs to a basis of $\Gamma_0(R)$, by 3.3.5. □

6.3 Radical and coradical

Let $\mathcal{R}$ be a root datum. Put

$$N = \{x \in M \mid (\alpha^*, x) = 0 \text{ for every } \alpha^* \in R^*\}, \quad N^* = M^* / (\mathcal{V}(R^*) \cap M^*).$$

Lemma 6.3.1. *Consider the canonical morphisms*

$$N \longrightarrow M, \quad M^* \longrightarrow N^*.$$

They are transposes of one another, and N^ is identified with the dual of N .*

Proof. This follows immediately from 1.2.5. □

Definition 6.3.2. The *coradical* of $\mathcal{R}$, denoted $\text{corad}(\mathcal{R})$, is the trivial root datum

$$\text{corad}(\mathcal{R}) = (N, N^*, \emptyset, \emptyset).$$

If

$$\mathcal{R}^0 = (M, M^*, \emptyset, \emptyset),$$

which is a trivial root datum, there is therefore a morphism

$$\text{corad}(\mathcal{R}) \longrightarrow \mathcal{R}^0.$$

Definition 6.3.3. The *radical* of $\mathcal{R}$, denoted $\text{rad}(\mathcal{R})$, is the trivial root datum ²⁷

$$\text{rad}(\mathcal{R}) = \text{corad}(\mathcal{R}^*)^*.$$

Thus there is a diagram

$$\begin{array}{ccc} \text{corad}(\mathcal{R}) & \xrightarrow{\quad\quad\quad} & \text{rad}(\mathcal{R}) \\ & \searrow \quad \quad \nearrow & \\ & \mathcal{R}^0 & \end{array}$$

whose transpose is the corresponding diagram for $\mathcal{R}^*$.

Lemma 6.3.4. *The canonical morphism*

$$u : \text{corad}(\mathcal{R}) \longrightarrow \text{rad}(\mathcal{R})$$

is an isogeny.

Definition 6.3.5. Put

$$N(\mathcal{R}) = K(u) = M / \left((\mathcal{V}(R) \cap M) + \bigcap_{\alpha \in R} \ker(\alpha^*) \right).$$

There is then a canonical pairing

$$N(\mathcal{R}) \times N(\mathcal{R}^*) \longrightarrow \mathbb{Q}/\mathbb{Z}.$$

Lemma 6.3.6. *One has*

$$\text{rgred}(\text{rad}(\mathcal{R})) = \text{rgred}(\text{corad}(\mathcal{R})) = \text{rgred}(\mathcal{R}) - \text{rgss}(\mathcal{R}),$$

and the following conditions are equivalent:

- (i) $\mathcal{R}$ is semisimple;
- (ii) $\text{rad}(\mathcal{R}) = 0$;
- (iii) $\text{corad}(\mathcal{R}) = 0$.

6.4 Products of root data

Definition 6.4.1. Let

$$\mathcal{R} = (M, M^*, R, R^*), \quad \mathcal{R}' = (M', M'^*, R', R'^*)$$

be two root data. The *product root datum* of $\mathcal{R}$ and $\mathcal{R}'$, denoted

$$\mathcal{R}'' = \mathcal{R} \times \mathcal{R}',$$

is the root datum (M'', M''^*, R'', R''^*) , where

$$\begin{aligned} M'' &= M \times M', & M''^* &= M^* \times M'^*, \\ R'' &= (R \times 0) \cup (0 \times R'), & R''^* &= (R^* \times 0) \cup (0 \times R'^*), \end{aligned}$$

and the map $\alpha \mapsto \alpha^*$ is the evident one.

²⁷Editors' note: Equivalently, if

$$P = \{y \in M^* \mid (y, \alpha) = 0 \text{ for every } \alpha \in R\}, \quad P^* = M / (\mathcal{V}(R) \cap M),$$

then $\text{rad}(\mathcal{R}) = (P^*, P, \emptyset, \emptyset)$.

Proposition 6.4.2. *Under the preceding hypotheses there are canonical isomorphisms*

$$\Gamma_0(R'') \simeq \Gamma_0(R) \times \Gamma_0(R'), \quad \mathcal{V}(R'') \simeq \mathcal{V}(R) \times \mathcal{V}(R'),$$

$$W(\mathcal{R}'') \simeq W(\mathcal{R}) \times W(\mathcal{R}'),$$

and similarly for the other associated constructions. Moreover,

$$\text{rgred}(\mathcal{R}'') = \text{rgred}(\mathcal{R}) + \text{rgred}(\mathcal{R}'), \quad \text{rgss}(\mathcal{R}'') = \text{rgss}(\mathcal{R}) + \text{rgss}(\mathcal{R}').$$

There is also a canonical isomorphism of root data

$$(\mathcal{R} \times \mathcal{R}')^* \simeq \mathcal{R}^* \times \mathcal{R}'^*.$$

The preceding definitions extend immediately to products of any finite number of factors.

Proposition 6.4.3. *Let*

$$\mathcal{R} = \mathcal{R}_1 \times \mathcal{R}_2 \times \cdots \times \mathcal{R}_n$$

be a product of root data. The following conditions are equivalent:

- (i) $\mathcal{R}$ is semisimple, respectively simply connected, respectively adjoint, respectively reduced;
- (ii) every $\mathcal{R}_i$ is semisimple, respectively simply connected, respectively adjoint, respectively reduced.

Consider the following special case. Let $\mathcal{R}_0$ be a trivial root datum and $\mathcal{R}_1$ a semisimple root datum. There is then a commutative diagram

$$\begin{array}{ccc} & \mathcal{R}_0 \times \mathcal{R}_1 & \\ \nearrow & & \searrow \\ \mathcal{R}_1 & \xrightarrow{\text{id}} & \mathcal{R}_1 \end{array}$$

Lemma 6.4.4. *There are canonical isomorphisms*

$$\begin{array}{ccc} \text{corad}(\mathcal{R}_0 \times \mathcal{R}_1) & \xrightarrow{\sim} & \text{rad}(\mathcal{R}_0 \times \mathcal{R}_1) \\ & \searrow \sim & \nearrow \sim \\ & \mathcal{R}_0 & \end{array}$$

In particular,

$$N(\mathcal{R}_0 \times \mathcal{R}_1) = 0.$$

We shall see later that, conversely, if $N(\mathcal{R}) = 0$, then $\mathcal{R}$ is the product of a semisimple root datum and a trivial root datum.

6.5 Induced and coinduced root data

Let $\mathcal{R} = (M, M^*, R, R^*)$ be a root datum, and let $N \subset M$ be a subgroup containing the roots, that is,

$$\Gamma_0(R) \subset N \subset M.$$

The canonical linear map $i_N : N \rightarrow M$ gives, by transposition, a linear map

$${}^t i_N : M^* \longrightarrow N^*.$$

Put

$$R_N = R, \quad R_N^* = {}^t i_N(R^*).$$

Lemma 6.5.1. *The quadruple*

$$\mathcal{R}_N = (N, N^*, R_N, R_N^*)$$

is a root datum, and i_N is a morphism.

Proof. First, ${}^t i_N$ induces an isomorphism from R^* onto R_N^* . Indeed, if $\alpha, \beta \in R$ and

$${}^t i_N(\alpha^*) = {}^t i_N(\beta^*),$$

then

$$(\alpha^*, x) = (\beta^*, x)$$

for every $x \in N$, in particular for $x \in R$. Hence $\alpha = \beta$ by 1.1.4. The rest follows without difficulty. $\square$

Definition 6.5.2. The root datum $\mathcal{R}_N$ is called the root datum *induced by $\mathcal{R}$ on N* .

Lemma 6.5.3. *Let $f : \mathcal{R}' \rightarrow \mathcal{R}$ be a morphism, and put*

$$N = f(M') \subset M.$$

Then f factors uniquely through i_N .

In particular, up to isomorphism, the isogenies $\mathcal{R}' \rightarrow \mathcal{R}$ correspond bijectively to the finite-index subgroups of

$$M/\Gamma_0(R).$$

This sharpens 6.2.7.

Now let $N^* \subset M^*$ be a subgroup containing R^* . Define the root datum *coinduced by $\mathcal{R}$ on N^** by

$$\mathcal{R}^{N^*} = (\mathcal{R}_{N^*}^*)^*.$$

There is a canonical morphism

$$p^{N^*} : \mathcal{R} \longrightarrow \mathcal{R}^{N^*}.$$

Lemma 6.5.4. *Let $f : \mathcal{R}' \rightarrow \mathcal{R}$ be a morphism. There exist subgroups $N \subset M$ and $N'^* \subset M'^*$ such that f factors as*

$$\begin{array}{ccc} \mathcal{R}' & \xrightarrow{f} & \mathcal{R} \\ p^{N'^*} \downarrow & & \uparrow i_N \\ \mathcal{R}'^{N'^*} & \xrightarrow[\sim]{f_0} & \mathcal{R}_N \end{array}$$

where f_0 is an isomorphism.

Proof. Take $N = f(M')$, as in 6.5.3. The resulting morphism $M' \rightarrow N$ is surjective, so its transpose is injective. Take its image for N'^* . $\square$

We now treat some special cases. If $N = \Gamma_0(R)$, write

$$\mathcal{R}_N = \text{ad}(\mathcal{R}).$$

If $N = \mathcal{V}(R) \cap M$, write

$$\mathcal{R}_N = \text{ss}(\mathcal{R}).$$

Thus there is a diagram

$$\text{ad}(\mathcal{R}) \longrightarrow \text{ss}(\mathcal{R}) \longrightarrow \mathcal{R}.$$

Put

$$\text{der}(\mathcal{R}) = \text{ss}(\mathcal{R}^*)^*, \quad \text{sc}(\mathcal{R}) = \text{ad}(\mathcal{R}^*)^*.$$

By duality there is a diagram

$$\mathcal{R} \longrightarrow \text{der}(\mathcal{R}) \longrightarrow \text{sc}(\mathcal{R}).$$

Proposition 6.5.5. (i) *In the first row of the diagram*

$$\begin{array}{ccccccc} \text{ad}(\mathcal{R}) & \longrightarrow & \text{ss}(\mathcal{R}) & \longrightarrow & \text{der}(\mathcal{R}) & \longrightarrow & \text{sc}(\mathcal{R}) \\ & & & \searrow & & \nearrow & \\ & & & \mathcal{R} & & & \end{array}$$

the four root data are semisimple and the three morphisms are isogenies.

(ii) *The datum $\text{ad}(\mathcal{R})$ is adjoint, and $\mathcal{R}$ is adjoint if and only if $\text{ad}(\mathcal{R}) \rightarrow \mathcal{R}$ is an isomorphism.*

(iii) *The datum $\text{sc}(\mathcal{R})$ is simply connected, and $\mathcal{R}$ is simply connected if and only if $\mathcal{R} \rightarrow \text{sc}(\mathcal{R})$ is an isomorphism.*

(iv) *The following conditions are equivalent:*

- (a) $\mathcal{R}$ is semisimple;
- (b) $\text{ss}(\mathcal{R}) \rightarrow \mathcal{R}$ is an isomorphism;
- (c) $\mathcal{R} \rightarrow \text{der}(\mathcal{R})$ is an isomorphism.

Consider the morphism

$$\text{ss}(\mathcal{R}) \longrightarrow \text{der}(\mathcal{R}).$$

The definitions immediately give the following result.

Lemma 6.5.6. *Let*

$$h : \text{ss}(\mathcal{R}) \longrightarrow \text{der}(\mathcal{R})$$

be the canonical isogeny. Then

$$K(h) \simeq N(\mathcal{R}).$$

Consider other special cases of induced root data. Put

$$N = \{x \in M \mid (\alpha^*, x) = 0 \text{ for every } \alpha \in R\} \times \Gamma_0(R).$$

The sum is direct by 1.2.5. It follows that $\mathcal{R}_N$ is identified with

$$\text{ad}(\mathcal{R}) \times \text{corad}(\mathcal{R}).$$

One may replace $\Gamma_0(R)$ by $\mathcal{V}(R) \cap M$, and then dualize both constructions. This gives the following diagram of root data:

$$\begin{array}{ccccccc}
 \text{ad}(\mathcal{R}) & \longrightarrow & \text{ss}(\mathcal{R}) & \longrightarrow & \text{der}(\mathcal{R}) & \longrightarrow & \text{sc}(\mathcal{R}) \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 \text{ad}(\mathcal{R}) \times \text{corad}(\mathcal{R}) & \longrightarrow & \text{ss}(\mathcal{R}) \times \text{corad}(\mathcal{R}) & \longrightarrow & \mathcal{R} & \longrightarrow & \text{der}(\mathcal{R}) \times \text{rad}(\mathcal{R}) \longrightarrow \text{sc}(\mathcal{R}) \times \text{rad}(\mathcal{R}) \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 \text{ad}(\mathcal{R}) & \longrightarrow & \text{ss}(\mathcal{R}) & \longrightarrow & \text{der}(\mathcal{R}) & \longrightarrow & \text{sc}(\mathcal{R})
 \end{array}$$

It is commutative, as is checked at once, and is self-dual in the evident sense. The horizontal morphisms are isogenies, and the composites of the vertical arrows are the identity.

Lemma 6.5.7. *Let h_1 and h_2 be the canonical isogenies*

$$\text{ss}(\mathcal{R}) \times \text{corad}(\mathcal{R}) \xrightarrow{h_1} \mathcal{R} \xrightarrow{h_2} \text{der}(\mathcal{R}) \times \text{rad}(\mathcal{R}).$$

Then

$$K(h_1) \simeq K(h_2) \simeq N(\mathcal{R}).$$

Proof. This is immediate from the definitions. □

Corollary 6.5.8. *For a root datum $\mathcal{R}$, the following conditions are equivalent:*

- (i) $N(\mathcal{R}) = 0$, that is, $\text{corad}(\mathcal{R}) \rightarrow \text{rad}(\mathcal{R})$ is an isomorphism;
- (ii) the canonical map $\text{ss}(\mathcal{R}) \rightarrow \text{der}(\mathcal{R})$ is an isomorphism;
- (iii) h_1 is an isomorphism;
- (iv) h_2 is an isomorphism;
- (v) $\mathcal{R}$ is the product of a semisimple root datum and a trivial root datum.

Corollary 6.5.9. *For every root datum $\mathcal{R}$, there are isogenies*

$$\text{ad}(\mathcal{R}) \times \mathcal{R}_0 \longrightarrow \mathcal{R} \longrightarrow \text{sc}(\mathcal{R}) \times \mathcal{R}_0,$$

where $\mathcal{R}_0$ is the trivial root datum of rank $\text{rgred}(\mathcal{R}) - \text{rgss}(\mathcal{R})$.

Lemma 6.5.10. *Let $\mathcal{R}$ be a root datum, let Δ be a system of simple roots, and let $\Delta' \subset \Delta$. Consider the root datum*

$$\mathcal{R}_{\Delta'} = (M, M^*, R_{\Delta'}, R_{\Delta'}^*)$$

of 3.4.7.

- (i) If $\mathcal{R}$ is simply connected, then $\text{der}(\mathcal{R}_{\Delta'})$ is simply connected.
- (ii) If $\mathcal{R}$ is adjoint, then $\text{ss}(\mathcal{R}_{\Delta'})$ is adjoint.

Proof. The two assertions are equivalent by duality. The second reduces to

$$M \cap \mathcal{V}(R_{\Delta'}) = \Gamma_0(R_{\Delta'}).$$

When $M = \Gamma_0(R)$, both sides are the subgroup of M generated by Δ' . □

6.6 Weights

Definition 6.6.1. For a root datum $\mathcal{R}$, put ²⁸

$$\Lambda(\mathcal{R}) = \{x \in \mathcal{V}(R) \mid (\alpha^*, x) \in \mathbb{Z} \text{ for every } \alpha^* \in R^*\}.$$

The elements of $\Lambda(\mathcal{R})$ are called the *weights* of $\mathcal{R}$. The weights of $\mathcal{R}^*$ are called the *coweights* of $\mathcal{R}$.

One has

$$\Gamma_0(R) \subset \Lambda(\mathcal{R}),$$

and $\Lambda(\mathcal{R})$ is stable under $W(\mathcal{R})$.

Lemma 6.6.2. *The bilinear map*

$$V^* \times V \longrightarrow \mathbb{Q}$$

induces a duality

$$\Gamma_0(R^*) \times \Lambda(\mathcal{R}) \longrightarrow \mathbb{Z}.$$

Proof. Immediate. □

Corollary 6.6.3. *Let*

$$\Delta^* = (\alpha_1^*, \dots, \alpha_n^*)$$

be a system of simple coroots, and let $p_i \in \mathcal{V}(R)$, $i = 1, \dots, n$, be defined by

$$(\alpha_i^*, p_j) = \delta_{ij}.$$

Thus

$$s_{\alpha_i}(p_i) = p_i - \alpha_i, \quad s_{\alpha_i}(p_j) = p_j \quad (i \neq j).^{29}$$

Then $\Lambda(\mathcal{R})$ is the free abelian group generated by the p_i .

The p_i are called the *fundamental weights* corresponding to the system of simple coroots Δ^* .

Corollary 6.6.4. *For every $\alpha^* \in \Delta^*$,*

$$\left(\alpha^*, \sum_i p_i \right) = 1.$$

Consequently,

$$\sum_i p_i = \rho_{R_+}, \quad R_+ = \mathcal{P}(\text{ind}(\Delta)),$$

in the notation of 3.5.1.

²⁸Editors' note: The notation $\Gamma(\mathcal{R})$ has been replaced by $\Lambda(\mathcal{R})$ to avoid any possible confusion with $\Gamma_0(R)$.

²⁹Caution: when the datum is not reduced, the α_i need not form a system of simple roots.

Corollary 6.6.5. *For every $x \in \mathcal{V}(R)$,*

$$x = \sum_i (\alpha_i^*, x) p_i.$$

Since

$$R^* \subset \Gamma_0(R^*), \quad R \subset \Lambda(\mathcal{R}),$$

the quadruple

$$(\Lambda(\mathcal{R}), \Gamma_0(R^*), R, R^*)$$

is a root datum.

Corollary 6.6.6. *The canonical morphism*

$$\Gamma_0(R^*) \longrightarrow M^*$$

is the transpose of

$$x \longmapsto \sum_i (\alpha_i^*, x) p_i.$$

This map defines a morphism of root data, and there is a commutative diagram

$$\begin{array}{ccc} \mathcal{R} & \xrightarrow{\quad} & \text{sc}(\mathcal{R}) \\ & \searrow & \swarrow \wr \\ & (\Lambda(\mathcal{R}), \Gamma_0(R^*), R, R^*) & \end{array}$$

Thus we have an explicit description of $\text{sc}(\mathcal{R})$ in terms of the weights of $\mathcal{R}$. Likewise, one obtains a commutative diagram

$$\begin{array}{ccc} \text{ad}(\mathcal{R}) & \xrightarrow{\quad} & \mathcal{R} \\ \downarrow \wr & & \swarrow \\ (\Gamma_0(R), \Lambda(\mathcal{R}^*), R, R^*) & & \end{array}$$

Corollary 6.6.7. *The datum $\mathcal{R}$ is simply connected if and only if*

$$M = \Lambda(\mathcal{R}).$$

Remark 6.6.8. One has

$$\Lambda(\mathcal{R}) \cap M = \mathcal{V}(R) \cap M.$$

Consequently, $\mathcal{R}$ is semisimple if and only if

$$M \subset \Lambda(\mathcal{R}).$$

The results of §6.5 also give:

Corollary 6.6.9. *The datum $\mathcal{R}$ is the product of a simply connected datum and a trivial datum if and only if*

$$M \supset \Lambda(\mathcal{R}).$$

Consider now the canonical isogeny

$$f : \mathrm{ad}(\mathcal{R}) \longrightarrow \mathrm{sc}(\mathcal{R}),$$

and put $Z(\mathcal{R}) = K(f)$. One has

$$Z(\mathcal{R}) \simeq Z(\mathrm{sc} \mathcal{R}) \simeq Z(\mathrm{ad} \mathcal{R}).$$

Corollary 6.6.10. *There is a canonical isomorphism*

$$Z(\mathcal{R}) = \Lambda(\mathcal{R})/\Gamma_0(R).$$

More precisely, there is an exact sequence of $W(\mathcal{R})$ -modules

$$0 \longrightarrow \Gamma_0(R) \longrightarrow \Lambda(\mathcal{R}) \longrightarrow Z(\mathcal{R}) \longrightarrow 0.$$

Corollary 6.6.11. *There is a canonical pairing*

$$Z(\mathcal{R}^*) \times Z(\mathcal{R}) \longrightarrow \mathbb{Q}/\mathbb{Z}$$

which puts these groups in duality.

Remark 6.6.12. One has

$$Z(\mathcal{R} \times \mathcal{R}') \simeq Z(\mathcal{R}) \times Z(\mathcal{R}').$$

In particular, consider simply connected data $\mathcal{R}_i$, $i = 1, 2, \dots, n$, and a trivial datum $\mathcal{R}_0$. Put

$$\mathcal{R} = \mathcal{R}_0 \times \mathcal{R}_1 \times \dots \times \mathcal{R}_n.$$

If

$$\mathcal{R} = (M, M^*, R, R^*), \quad \mathcal{R}_0 = (M_0, M_0^*, \emptyset, \emptyset),$$

then

$$M/\Gamma_0(R) \simeq M_0 \times Z(\mathcal{R}_1) \times \dots \times Z(\mathcal{R}_n).$$

6.7 Automorphisms

By 6.1.4, an automorphism of $\mathcal{R}$ is an automorphism u of M such that

$$u(R) = R, \quad {}^t u(R^*) = R^*.$$

In particular, every $w \in W(\mathcal{R})$ defines an automorphism of $\mathcal{R}$.

Lemma 6.7.1. *The group $W(\mathcal{R})$ is a normal subgroup of $\mathrm{Aut}(\mathcal{R})$. More precisely, if $u \in \mathrm{Aut}(\mathcal{R})$ and $\alpha \in R$, then*

$$us_\alpha u^{-1} = s_{u(\alpha)}.$$

Proof. The proof is the same as that of 1.2.10. □

Proposition 6.7.2. *Let Δ be a system of simple roots, and put*

$$E_\Delta(\mathcal{R}) = \{u \in \mathrm{Aut}(\mathcal{R}) \mid u(\Delta) = \Delta\}.$$

Then $\mathrm{Aut}(\mathcal{R})$ is the semidirect product of $W(\mathcal{R})$ by $E_\Delta(\mathcal{R})$.

Proof. This follows at once from the fact that $W(\mathcal{R})$ acts simply transitively on the systems of simple roots and from the fact that, if Δ is a system of simple roots of $\mathcal{R}$, then $u(\Delta)$ is a system of simple roots for every automorphism u of $\mathcal{R}$. □

We shall later give a simpler description of $E_\Delta(\mathcal{R})$ in the case of reduced irreducible root data.³⁰

Definition 6.7.3. We denote by $\text{Aut}^s(\mathcal{R})$ ³¹ the set of $u \in \text{Aut}(\mathcal{R})$ for which the following diagram is commutative:

$$\begin{array}{ccc} \mathcal{R} & \xrightarrow{\quad u \quad} & \mathcal{R} \\ & \searrow & \swarrow \\ & \text{rad}(\mathcal{R}) & \end{array}$$

We write

$$E_\Delta^s(\mathcal{R}) = E_\Delta(\mathcal{R}) \cap \text{Aut}^s(\mathcal{R}).$$

Remark 6.7.4. If $u \in \text{Aut}(\mathcal{R})$, then $u \in \text{Aut}^s(\mathcal{R})$ if and only if

$$(u - \text{id})(M) \subset \mathcal{V}(R).$$

In particular,

$$W(\mathcal{R}) \subset \text{Aut}^s(\mathcal{R}).$$

Proposition 6.7.5. *For every system of simple roots Δ , the group $\text{Aut}^s(\mathcal{R})$ is the semidirect product of $W(\mathcal{R})$ by $E_\Delta^s(\mathcal{R})$.*

By functoriality, every automorphism of $\mathcal{R}$ induces an automorphism of $\text{ad}(\mathcal{R})$. Thus there is a canonical homomorphism

$$\text{Aut}(\mathcal{R}) \longrightarrow \text{Aut}(\text{ad}(\mathcal{R})).$$

Lemma 6.7.6. *The homomorphism*

$$\text{Aut}^s(\mathcal{R}) \longrightarrow \text{Aut}(\text{ad}(\mathcal{R}))$$

is injective.

Proof. Let u be an automorphism of M such that

$$(u - \text{id})(M) \subset \mathcal{V}(R)$$

and

$${}^t u(\alpha^*) = \alpha^* \quad (\alpha^* \in R^*).$$

For every $x \in M$, one has

$$(\alpha^*, u(x) - x) = ({}^t u(\alpha^*) - \alpha^*, x) = 0.$$

Hence $u(x) - x = 0$, by 1.2.5. □

Lemma 6.7.7. *The group $\text{Aut}^s(\mathcal{R})$ is finite.*

Proof. It suffices to prove that $\text{Aut}(\mathcal{R})$ is finite when $\mathcal{R}$ is adjoint. In that case M is generated by R , so every automorphism of $\mathcal{R}$ is determined by the permutation it induces on R . □

Remark 6.7.8. It is immediate that $\text{Aut}(\mathcal{R})$ (respectively $E_\Delta(\mathcal{R})$) is finite if and only if

$$\text{rgred}(\mathcal{R}) - \text{rgss}(\mathcal{R}) \leq 1.$$

³⁰Editors' note: When $\mathcal{R}$ is simply connected or adjoint, see 7.4.5.

³¹Editors' note: The superscript s is intended to suggest "semisimple."

6.8 p -morphisms of reduced root data

Throughout this subsection, p is a fixed positive integer.

Definition 6.8.1. Let

$$\mathcal{R} = (M, M^*, R, R^*), \quad \mathcal{R}' = (M', M'^*, R', R'^*)$$

be two reduced root data. A group homomorphism

$$f : M' \longrightarrow M$$

is called a p -morphism from $\mathcal{R}'$ to $\mathcal{R}$ if there exist a bijection

$$u : R \xrightarrow{\sim} R'$$

and a map

$$q : R \longrightarrow \{p^n \mid n \in \mathbb{N}\}$$

such that:

- (i) $f(u(\alpha)) = q(\alpha)\alpha$ for every $\alpha \in R$;
- (ii) ${}^t f(\alpha^*) = q(\alpha)u(\alpha)^*$ for every $\alpha \in R$.³²

Corollary 6.8.2. *A 1-morphism is precisely a morphism.*

Corollary 6.8.3. *The transpose of a p -morphism is a p -morphism.*

Lemma 6.8.4. *If $w \in W(\mathcal{R})$ and $\alpha \in R$, then*

$$q(w(\alpha)) = q(\alpha).$$

The map

$$s_\alpha \longmapsto s_{u(\alpha)}$$

extends to an isomorphism

$$u : W(\mathcal{R}) \xrightarrow{\sim} W(\mathcal{R}')$$

such that

$$u(w(\alpha)) = u(w)(u(\alpha)).$$

Proof. It suffices to prove, for $\alpha, \beta \in R$, that

$$u(s_\alpha(\beta)) = s_{u(\alpha)}u(\beta), \quad q(s_\alpha(\beta)) = q(\beta).$$

Successively, one has

$$\begin{aligned} f(s_{u(\alpha)}u(\beta)) &= f(u(\beta)) - (u(\alpha)^*, u(\beta))f(u(\alpha)) \\ &= q(\beta)\beta - q(\beta)q(\alpha)^{-1}(\alpha^*, \beta)q(\alpha)\alpha \\ &= q(\beta)(\beta - (\alpha^*, \beta)\alpha) \\ &= q(\beta)s_\alpha(\beta). \end{aligned}$$

³²Editors' note: These two conditions imply

$$q(\alpha)(u(\alpha)^*, u(\beta)) = q(\beta)(\alpha^*, \beta)$$

for all $\alpha, \beta \in R$.

If

$$\gamma = u^{-1}(s_{u(\alpha)}u(\beta)),$$

then

$$q(\gamma)\gamma = f(u(\gamma)) = q(\beta)s_\alpha(\beta).$$

The roots γ and $s_\alpha(\beta)$ are proportional over $\mathbb{Q}$, hence equal or opposite. Since $q(\gamma)$ and $q(\beta)$ are positive, it follows that

$$q(\gamma) = q(\beta), \quad \gamma = s_\alpha(\beta).$$

□

Definition 6.8.5. The numbers $q(\alpha)$ are called the *root exponents* of f .

Example 6.8.6. Let $\mathcal{R}$ be a reduced root datum and let $q = p^n$, $n \in \mathbb{N}$. Multiplication by q ,

$$M \longrightarrow M, \quad x \longmapsto qx,$$

is a p -morphism whose root exponents are all equal to q (and for which $u = \text{id}$). It is denoted

$$q : \mathcal{R} \longrightarrow \mathcal{R}.$$

Proposition 6.8.7. *In the notation of 6.8.1, u induces an isomorphism from the set of systems of simple roots (respectively positive roots) of R onto the corresponding set for R' .*

Proof. This follows immediately from 3.1.5 (respectively 3.2.1). □

7 Structure

7.1 Decomposition of a root datum

Proposition 7.1.1. *Let $\mathcal{R}$ be a root datum and let Δ be a system of simple roots.*

(i) *Let R' and R'' be two closed symmetric sets of roots forming a partition of R . If*

$$\Delta' = \Delta \cap R', \quad \Delta'' = \Delta \cap R'',$$

then

$$R' = R_{\Delta'}, \quad R'' = R_{\Delta''},$$

and every root of Δ' is orthogonal to every root of Δ'' .

(ii) *Let Δ' and Δ'' be two orthogonal subsets forming a partition of Δ . Then $R_{\Delta'}$ and $R_{\Delta''}$ form a partition of R .*

Proof. We first prove (i).

Lemma 7.1.2. *Under the hypotheses of (i), if α , β , and $\alpha + \beta$ are roots, then all three belong to R' , or all three belong to R'' .*

Suppose, for example, that $\alpha + \beta \in R'$. One cannot have $\alpha, \beta \in R''$, since R'' is closed. Suppose therefore that $\alpha \in R'$. Then $-\alpha \in R'$, and

$$\beta = (\beta + \alpha) - \alpha \in R'.$$

We now prove $R' = R_{\Delta'}$ by induction on the order of a positive root $\alpha \in R' \cap \mathcal{P}(\Delta)$. If $\text{ord}_{\Delta}(\alpha) = 1$, then

$$\alpha \in R' \cap \Delta = \Delta'.$$

If $\text{ord}_{\Delta}(\alpha) > 1$, there is $\beta \in \Delta$ such that $\alpha - \beta \in R$. By the lemma,

$$\beta \in \Delta', \quad \alpha - \beta \in R'.$$

Hence $\alpha - \beta \in R_{\Delta'}$ by induction, and finally

$$\alpha = (\alpha - \beta) + \beta \in R_{\Delta'}.$$

We next prove that Δ' and Δ'' are orthogonal. If $\alpha \in \Delta'$ and $\beta \in \Delta''$, then

$$(\beta^*, \alpha) \leq 0.$$

If $(\beta^*, \alpha) \neq 0$, then $\beta + \alpha$ is a root, contrary to the lemma.

We now prove (ii). If Δ' or Δ'' is empty, the assertion is immediate. Otherwise, if $R_{\Delta'}$ and $R_{\Delta''}$ do not form a partition of R , there is a root α of the form

$$\alpha = \sum_i m'_i \alpha'_i + \sum_j m''_j \alpha''_j, \quad m'_i, m''_j \in \mathbb{Z}_+,$$

where the α'_i (respectively α''_j) belong to Δ' (respectively Δ''). Applying 3.1.2, and interchanging Δ' and Δ'' if necessary, yields a relation

$$\delta = \gamma + \beta, \quad \gamma \in R_{\Delta'}, \quad \beta \in \Delta'', \quad \delta \in R.$$

But $(\beta^*, \gamma) = 0$, so $\gamma - \beta$ is also a root by 2.2.5, which is impossible. $\square$

Proposition 7.1.3. *Let $\mathcal{R}$ be a root datum. The following conditions are equivalent:*

- (i) *There is no nontrivial partition of R into two closed symmetric subsets.*
- (ii) *For some (respectively every) system of simple roots Δ of R , there is no partition of Δ into two nonempty orthogonal subsets.*
- (iii) *The natural representation of $W(\mathcal{R})$ on $\mathcal{V}(R)$ is irreducible.*
- (iv) *For every pair (α, β) of roots, there is a sequence of roots*

$$\alpha_0, \alpha_1, \dots, \alpha_n, \quad \alpha_0 = \alpha, \quad \alpha_n = \beta,$$

such that α_i and α_{i+1} are nonorthogonal for $i = 0, \dots, n-1$.

Proof. The equivalence (i) $\Leftrightarrow$ (ii) follows from 7.1.1, and (iv) $\Rightarrow$ (ii) is immediate. Conversely, if (ii) holds for Δ , then (iv) holds when $\alpha, \beta \in \Delta$. Every root is nonorthogonal to some simple root, by 3.1.1, for example. Moreover, (iii) $\Rightarrow$ (i): under the hypotheses of 7.1.1, the space $\mathcal{V}(R')$ is stable under $W(\mathcal{R})$. It remains to prove (i) $\Rightarrow$ (iii).

Let H be a vector subspace of $\mathcal{V}(R)$, stable under $W(\mathcal{R})$. For every $\alpha \in R$, the equality $s_{\alpha}(H) = H$ immediately gives

$$\alpha \in H \quad \text{or} \quad \alpha^* \in H^{\perp},$$

where $H^{\perp}$ denotes the orthogonal complement of H in $\mathcal{V}(R^*)$, which is in duality with $\mathcal{V}(R)$. Put

$$R' = \{\alpha \in R \mid \alpha \in H\}, \quad R'' = \{\alpha \in R \mid \alpha^* \in H^{\perp}\}.$$

This gives a partition of R into two closed symmetric subsets. $\square$

Definition 7.1.4. A root datum (respectively root system) satisfying the equivalent conditions of 7.1.3 and having nonzero semisimple rank is called *irreducible*.

Corollary 7.1.5. *For every root datum $\mathcal{R}$, there is a unique partition, up to order, of R into closed symmetric irreducible subsets.*

Corollary 7.1.6. *Every adjoint (respectively simply connected) root datum is a product of irreducible adjoint (respectively simply connected) root data.*

Proof. It suffices to consider the adjoint case. The assertion then follows from the fact that, under the hypotheses of 7.1.1,

$$\Gamma_0(R) = \Gamma_0(R') \times \Gamma_0(R'').$$

□

Corollary 7.1.7. *For every root datum (respectively reduced root datum) $\mathcal{R}$, there is an isogeny*

$$\mathcal{R} \longrightarrow \mathcal{R}',$$

where $\mathcal{R}'$ is a product of a trivial root datum and irreducible simply connected root data (respectively irreducible reduced simply connected root data).

7.2 Properties of irreducible root data

Definition 7.2.1. Let $\mathcal{R}$ be an irreducible root datum. For every $\alpha \in R$, put

$$\text{long}(\alpha) = \ell(\alpha)/\ell(\alpha_0),$$

where $\alpha_0 \in R$ is such that $\ell(\alpha_0)$ is minimal. The number $\text{long}(\alpha)$ is called the *length* of α .

Lemma 7.2.2. *Let $\mathcal{R}$ be an irreducible root datum. The Weyl group acts transitively on the set of roots of a given length.*

Proof. Let $\alpha, \beta \in R$. Since the representation of $W(\mathcal{R})$ on $\mathcal{V}(R)$ is irreducible, α cannot be orthogonal to all $w(\beta)$, $w \in W(\mathcal{R})$. Hence some $w(\beta)$ is nonorthogonal to α . Since

$$\ell(w(\beta)) = \ell(\beta),$$

the assertion follows from 2.3.2. □

Lemma 7.2.3. *If $\mathcal{R}$ is irreducible and reduced, then*

$$\text{long}(R) \in \{\{1\}, \{1, 2\}, \{1, 3\}\}.$$

Proof. By the observation used above, for every $\alpha, \beta \in R$ there is $w \in W(\mathcal{R})$ such that $w(\beta)$ is nonorthogonal to α . Therefore, by 2.3.1,

$$\frac{\ell(\alpha)}{\ell(\beta)} \in \left\{1, 2, 3, \frac{1}{2}, \frac{1}{3}\right\}.$$

Thus $\text{long}(\alpha)$ is 1, 2, or 3. But lengths 2 and 3 cannot both occur, since their ratio $2/3$ is not in the displayed set. □

Remark 7.2.4. Reasoning similarly gives the following result. If $\mathcal{R}$ is irreducible and nonreduced, with $\text{rgss}(\mathcal{R}) > 1$, then

$$\text{long}(R) = \{1, 2, 4\}.$$

If $R_i = \text{long}^{-1}(i)$, then

$$\text{ind}(R) = R_1 \cup R_2, \quad R_4 = 2R_1,$$

and two nonproportional roots of R_1 are orthogonal. Conversely, suppose that R is an irreducible reduced root system such that $\text{long}(R) = \{1, 2\}$. Put $R_i = \text{long}^{-1}(i)$, and suppose that two nonproportional roots of R_1 are orthogonal. Then $R \cup 2R_1$ is irreducible and nonreduced, and

$$\text{ind}(R \cup 2R_1) = R.$$

Lemma 7.2.5. *If $\mathcal{R}$ is an irreducible root datum, then $\mathcal{R}^*$ is irreducible as well, and the product*

$$\text{long}(\alpha) \text{long}(\alpha^*)$$

is constant as α runs through R .

Proof. This follows immediately from 7.1.3 (iv) and 2.2.6. $\square$

Definition 7.2.6. Let $\mathcal{R}$ be any root datum. The *length* of $\alpha \in R$, denoted $\text{long}(\alpha)$, is the length of α in its irreducible component.

Lemma 7.2.7. *There is a unique group homomorphism*

$$u : \Gamma_0(R) \longrightarrow \Gamma_0(R^*)$$

such that

$$u(\alpha) = \text{long}(\alpha)\alpha^* \quad (\alpha \in R).$$

Proof. By 3.5.5, it is enough to verify that, whenever $\alpha, \beta, \alpha + \beta \in R$,

$$\text{long}(\alpha)\alpha^* + \text{long}(\beta)\beta^* = \text{long}(\alpha + \beta)(\alpha + \beta)^*.$$

But $\alpha, \beta, \alpha + \beta$ belong to the same irreducible component of R , by 7.1.2; the assertion is therefore reduced to 1.2.2. $\square$

Remark 7.2.8. Let u be as in 7.2.7. For $\alpha, \beta \in R$,

$$(u(\alpha), \beta) = (u(\beta), \alpha).$$

Indeed, this amounts to

$$\text{long}(\alpha)(\alpha^*, \beta) = \text{long}(\beta)(\beta^*, \alpha),$$

which is clear when α and β are orthogonal. If they are not orthogonal, they lie in the same irreducible component of R , and the assertion reduces to 1.2.1, formula (9).

Remark 7.2.9. The symmetric bilinear form

$$(x, y) \longmapsto (u(x), y)$$

is positive and nondegenerate on $\Gamma_0(R)$.

Proof. Let R_i be the irreducible components of R . Then

$$\Gamma_0(R) = \prod_i \Gamma_0(R_i),$$

and the bilinear form $(u(x), y)$ is the product, on the groups $\Gamma_0(R_i)$, of the forms

$$2^{-1}\ell(\alpha_i)^{-1}(p(x), y),$$

where $\ell(\alpha_i)$ is the minimum of $\ell(\alpha)$ for $\alpha \in R_i$. These symmetric bilinear forms are positive and nondegenerate by 1.2.6. $\square$

7.3 Cartan matrix

Let $\mathcal{R}$ be a root datum. If Δ is a system of simple roots, the *Cartan matrix* of $\mathcal{R}$ relative to Δ is the square matrix indexed by Δ and defined by

$$a_{\alpha,\beta} = (\alpha^*, \beta), \quad \alpha, \beta \in \Delta.$$

If Δ' is another system of simple roots and $w \in W(\mathcal{R})$ satisfies $w(\Delta) = \Delta'$, then

$$(w(\alpha)^*, w(\beta)) = (\alpha^*, \beta).$$

Thus the Cartan matrix relative to Δ' is obtained from that relative to Δ by the isomorphism $\Delta \rightarrow \Delta'$ of index sets defined by w . Consequently, up to isomorphism of its index set, the Cartan matrix depends only on $\mathcal{R}$.

Proposition 7.3.1. *The Cartan matrix has the following properties.*

- (i) $a_{\alpha,\alpha} = 2$, and $a_{\alpha,\beta} \leq 0$ for $\alpha \neq \beta$.
- (ii) $a_{\alpha,\beta} = 0$ implies $a_{\beta,\alpha} = 0$.
- (iii) There are strictly positive integers $m_\alpha = \text{long}(\alpha)$ such that the matrix

$$(m_\alpha a_{\alpha,\beta})$$

is symmetric, positive, and nondegenerate.

- (iv) The principal minors of $(a_{\alpha,\beta})_{\alpha,\beta \in \Delta}$, namely

$$\det(a_{\alpha,\beta})_{\alpha,\beta \in \Delta'} \quad (\Delta' \subset \Delta),$$

are strictly positive.

- (v) One has

$$s_\alpha(\beta) = \beta - a_{\alpha,\beta}\alpha, \quad s_\alpha(\beta^*) = \beta^* - a_{\beta,\alpha}\alpha^*.$$

Proof. Part (v) is a definition; (i) follows from 3.2.11, (ii) from 2.2.2, and (iii) from 7.2.9. Part (iv) follows immediately from (iii) and

$$\det(m_\alpha a_{\alpha,\beta})_{\alpha,\beta \in \Delta'} = \left(\prod_{\alpha \in \Delta'} m_\alpha \right) \det(a_{\alpha,\beta})_{\alpha,\beta \in \Delta'}.$$

□

Proposition 7.3.2. *Let $\mathcal{R}$ and $\mathcal{R}'$ be two reduced simply connected (respectively adjoint) root data, let Δ and Δ' be systems of simple roots of $\mathcal{R}$ and $\mathcal{R}'$, and let $u : \Delta \rightarrow \Delta'$ be an isomorphism. Denote the corresponding Cartan matrices by $(a_{\alpha,\beta})$ and $(a'_{\alpha',\beta'})$, and suppose that*

$$a'_{u(\alpha),u(\beta)} = a_{\alpha,\beta}.$$

Then there is a unique isomorphism from $\mathcal{R}$ onto $\mathcal{R}'$ that induces u on Δ .

Proof. It is enough to give the proof in the adjoint case. Then $M = \Gamma_0(R)$ and $M' = \Gamma_0(R')$ are the free abelian groups generated by Δ and Δ' . Hence there is a unique group isomorphism $M \rightarrow M'$ inducing u on Δ ; denote it again by u .

We show that $u(R) \subset R'$. Every root α of R can be written

$$\alpha = s_{\alpha_1} \cdots s_{\alpha_n}(\alpha_{n+1}), \quad \alpha_i \in \Delta.$$

By the hypothesis on u and the relations 7.3.1 (v),

$$u(\alpha) = s_{u(\alpha_1)} \cdots s_{u(\alpha_n)}(u(\alpha_{n+1})).$$

It remains to prove that ${}^t u(R^*) \subset R^*$. This follows from the fact that the elements of M^* (respectively M'^*) are determined by their duality with R or Δ (respectively R' or Δ'), by 1.2.5. $\square$

Corollary 7.3.3. *A reduced simply connected or adjoint root datum is determined up to isomorphism by its Cartan matrix.*

Corollary 7.3.4. *Let $\mathcal{R}$ be a reduced simply connected (respectively adjoint) root datum and let Δ be a system of simple roots. The group $E_\Delta(\mathcal{R})$ is identified with the group of automorphisms of the set Δ that preserve the Cartan matrix.*

Remark 7.3.5. The existence of a root datum corresponding to a given Cartan matrix satisfying, for example, (i), (ii), and (iv) is not easily settled directly without using the classification.

7.4 Dynkin diagrams

Definition 7.4.1. A *Dynkin-diagram structure* on a finite set Δ (the word “scheme” being banished for obvious reasons) consists of a set of pairs of distinct elements of Δ , called *linked pairs*, together with a map

$$\Delta \longrightarrow \{1, 2, 3\}.$$

The notion of isomorphism between such structures is evident.

Definition 7.4.2. Let $\mathcal{R}$ be a root datum and let Δ be a system of simple roots. The *Dynkin diagram* of $\mathcal{R}$ relative to Δ is the set Δ , in which two simple roots are linked if and only if they are nonorthogonal, and each root is assigned its length.

Proposition 7.4.3. *The Dynkin diagram and the Cartan matrix determine one another uniquely.*

Proof. The equivalence

$$\alpha \text{ and } \beta \text{ are not linked} \iff a_{\alpha,\beta} = 0$$

and the relation

$$\text{long}(\alpha)a_{\alpha,\beta} = \text{long}(\beta)a_{\beta,\alpha},$$

with $\inf \text{long}(\alpha) = 1$ in each connected component of the diagram, determine the $a_{\alpha,\beta}$ from the links and lengths, and conversely. The details are left to the reader. $\square$

Corollary 7.4.4. *A reduced simply connected or adjoint root datum is determined by its Dynkin diagram.*

Corollary 7.4.5. *Let $\mathcal{R}$ be a reduced simply connected or adjoint root datum and let Δ be a system of simple roots. The group $E_\Delta(\mathcal{R})$ is identified with the group of automorphisms of the Dynkin diagram of $\mathcal{R}$ relative to Δ ; that is, with the group of permutations of Δ preserving lengths and links.*

Remark 7.4.6. The usual method^{33 34} classifies the connected Dynkin diagrams and shows that each actually corresponds to an irreducible reduced simply connected root datum. One obtains the familiar types:

$$\begin{array}{ll}
 \begin{array}{c} 1 \quad 1 \quad 1 \\ \bullet \text{---} \bullet \text{---} \bullet \end{array} \quad \dots \quad \begin{array}{c} 1 \quad 1 \quad 1 \\ \bullet \text{---} \bullet \text{---} \bullet \end{array} & A_n, \quad n \geq 1. \\
 \begin{array}{c} 2 \quad 2 \quad 2 \\ \bullet \text{---} \bullet \text{---} \bullet \end{array} \quad \dots \quad \begin{array}{c} 2 \quad 2 \quad 1 \\ \bullet \text{---} \bullet \text{---} \bullet \end{array} & B_n, \quad n \geq 2. \\
 \begin{array}{c} 1 \quad 1 \quad 1 \\ \bullet \text{---} \bullet \text{---} \bullet \end{array} \quad \dots \quad \begin{array}{c} 1 \quad 1 \quad 2 \\ \bullet \text{---} \bullet \text{---} \bullet \end{array} & C_n, \quad n \geq 3. \\
 \begin{array}{c} 1 \quad 1 \quad 1 \\ \bullet \text{---} \bullet \text{---} \bullet \end{array} \quad \dots \quad \begin{array}{c} 1 \\ \swarrow \bullet \\ \searrow \bullet \end{array} & D_n, \quad n \geq 4. \\
 \begin{array}{c} 1 \quad 1 \quad \dots \quad 1 \quad 1 \quad 1 \\ \bullet \text{---} \bullet \quad \dots \quad \bullet \text{---} \bullet \text{---} \bullet \\ \quad \quad \quad \downarrow \bullet \\ \quad \quad \quad 1 \end{array} & E_n, \quad n = 6, 7, 8. \\
 \begin{array}{c} 1 \quad 1 \quad 2 \quad 2 \\ \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \end{array} & F_4. \\
 \begin{array}{c} 1 \quad 3 \\ \bullet \text{---} \bullet \end{array} & G_2.
 \end{array}$$

By 7.4.5, the corresponding group $E_\Delta(\mathcal{R})$ is obtained at once:

$$\begin{aligned}
 E_\Delta(\mathcal{R}) &= \{e\}, & A_1, B_n, C_n, E_7, E_8, F_4, G_2, \\
 E_\Delta(\mathcal{R}) &= \mathbb{Z}/2\mathbb{Z}, & A_n \ (n \geq 2), D_n \ (n \geq 5), E_6, \\
 E_\Delta(\mathcal{R}) &= \mathfrak{S}_3, & D_4.
 \end{aligned}$$

7.5 Complements on p -morphisms

Let $f : \mathcal{R} \rightarrow \mathcal{R}'$ be a p -morphism (cf. 6.8). It is clear from the definitions that the bijection $u : R \xrightarrow{\sim} R'$ associated with f makes the systems of simple roots, the systems of positive roots, the irreducible components, and so forth of R and R' correspond. We may therefore suppose, for simplicity, that R and R' are irreducible.

Lemma 7.5.1. *If R and R' are irreducible, there is a $k \in \mathbb{Q}$ such that, for every $\alpha \in R$,*

$$k \operatorname{long}(u(\alpha)) = q(\alpha)^2 \operatorname{long}(\alpha).$$

Proof. One has

$$\operatorname{long}(\alpha)(\alpha^*, \beta) = \operatorname{long}(\beta)(\beta^*, \alpha)$$

and, similarly,

$$\operatorname{long}(u(\alpha))(u(\alpha)^*, u(\beta)) = \operatorname{long}(u(\beta))(u(\beta)^*, u(\alpha)).$$

³³See Bourbaki, *Groupes et algèbres de Lie*, Chapter VI, no. 4.2, or the Sophus Lie Seminar.

³⁴Editors' note: for a slightly different proof, see also [De80].

It follows immediately that, for nonorthogonal α, β ,

$$\frac{q(\alpha)^2 \operatorname{long}(\alpha)}{\operatorname{long}(u(\alpha))} = \frac{q(\beta)^2 \operatorname{long}(\beta)}{\operatorname{long}(u(\beta))}.$$

The conclusion then follows from 7.1.3 (iv). $\square$

Remark 7.5.2. It follows from 7.2.2 and 6.8.4 that $q(\alpha)$ depends only on $\operatorname{long}(\alpha)$. One then sees readily that, if $q(\alpha)$ is not constant, the product $q(\alpha) \operatorname{long}(\alpha)$ is constant; hence $p = 2$ or 3 . Inspection of the preceding diagrams gives the four possible cases (the same letter denotes a Dynkin diagram and the corresponding reduced simply connected root datum):

$$\begin{aligned} p = 2, \quad & B_n \xrightarrow{f_1} C_n, \quad C_n \xrightarrow{f_2} B_n, \quad (C_2 = B_2), \\ p = 2, \quad & F_4 \xrightarrow{g} F_4, \\ p = 3, \quad & G_2 \xrightarrow{h} G_2. \end{aligned}$$

The reader will observe that $f_1 \circ f_2$, $f_2 \circ f_1$, $g \circ g$, and $h \circ h$ are p -morphisms of the form described in 6.8.6.

7.5.3. The preceding description shows immediately that, if $\mathcal{R}$ and $\mathcal{R}'$ are two reduced root data of semisimple rank at most 2 and there is a p -morphism from $\mathcal{R}'$ to $\mathcal{R}$, then $\mathcal{R}$ and $\mathcal{R}'$ have the same type. More precisely, one has the following table.

Notation. Let $f : \mathcal{R}' \rightarrow \mathcal{R}$ be a p -morphism. The symbols q and q_1 denote arbitrary positive powers of p . For rank-two systems we use the notation of Section 4, with α, β the simple roots and $\ell(\alpha) \leq \ell(\beta)$.

Type	p	values of f	values of ${}^t f$
Trivial	arbitrary	—	—
A_1	arbitrary	$f(\alpha') = q\alpha$	${}^t f(\alpha^*) = q\alpha'^*$
$A_1 \times A_1$	arbitrary	$f(\alpha') = q\alpha$ $f(\beta') = q_1\beta$	${}^t f(\alpha^*) = q\alpha'^*$ ${}^t f(\beta^*) = q_1\beta'^*$
A_2, B_2, G_2	arbitrary	$f(\alpha') = q\alpha$ $f(\beta') = q\beta$	${}^t f(\alpha^*) = q\alpha'^*$ ${}^t f(\beta^*) = q\beta'^*$
B_2	$p = 2$	$f(\alpha') = q\beta$ $f(\beta') = 2q\alpha$	${}^t f(\alpha^*) = 2q\beta'^*$ ${}^t f(\beta^*) = q\alpha'^*$
G_2	$p = 3$	$f(\alpha') = q\beta$ $f(\beta') = 3q\alpha$	${}^t f(\alpha^*) = 3q\beta'^*$ ${}^t f(\beta^*) = q\alpha'^*$

Bibliography

- [BLie] N. Bourbaki, *Groupes et algèbres de Lie*, Chapters IV–VI, Hermann, 1968.
- [De80] M. Demazure, “A, B, C, D, E, F, etc.,” pp. 221–227 in *Séminaire sur les singularités des surfaces* (Palaiseau, 1976–1977), eds. M. Demazure, H. C. Pinkham, and B. Teissier, Lecture Notes in Mathematics **777**, Springer-Verlag, 1980.